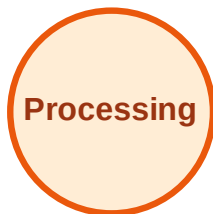


BFS Traversal

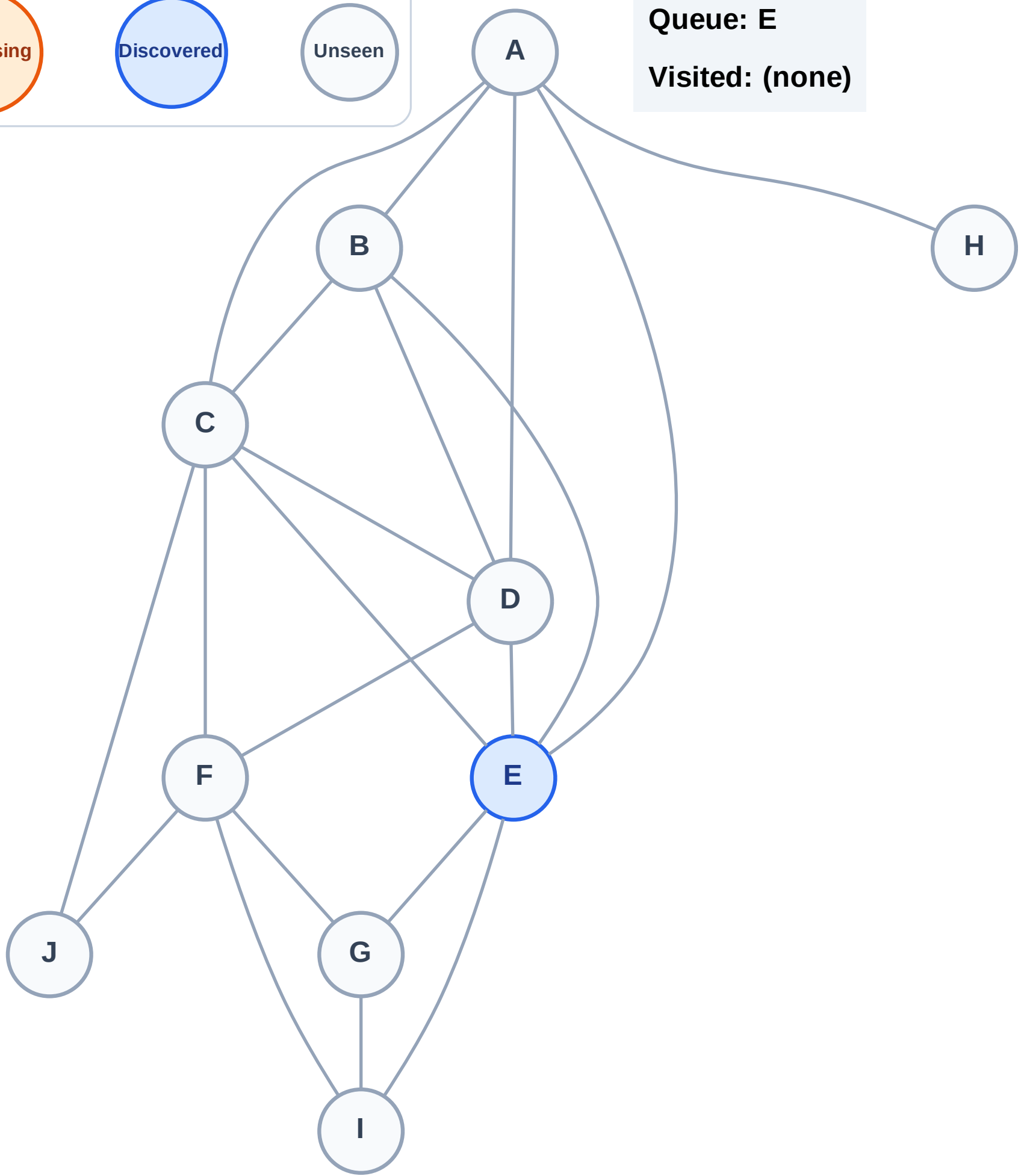
Step 1: Start at E

Legend



Queue: E

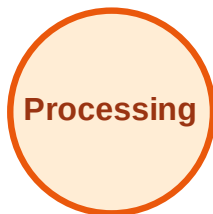
Visited: (none)



BFS Traversal

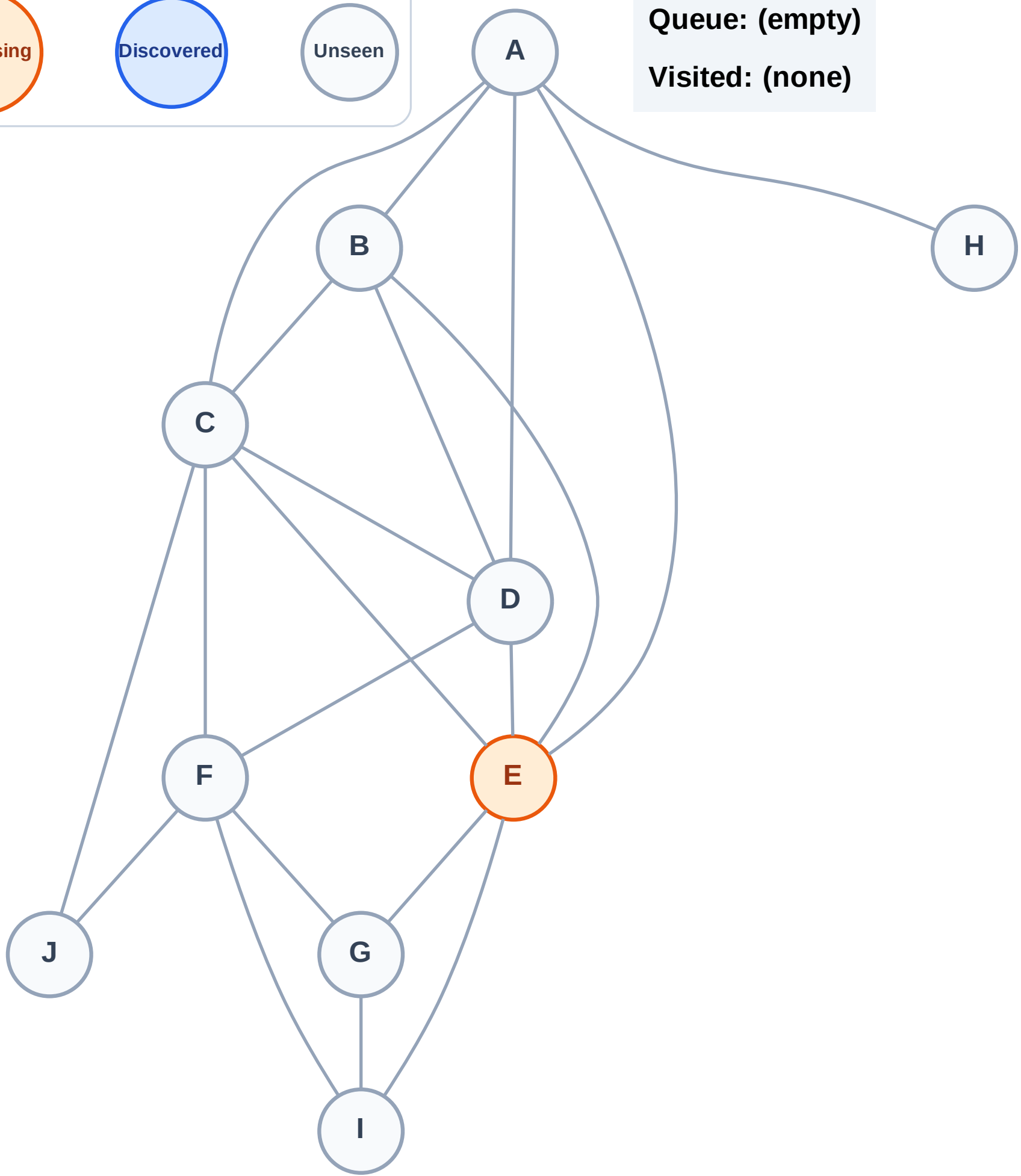
Step 2: Process node E

Legend



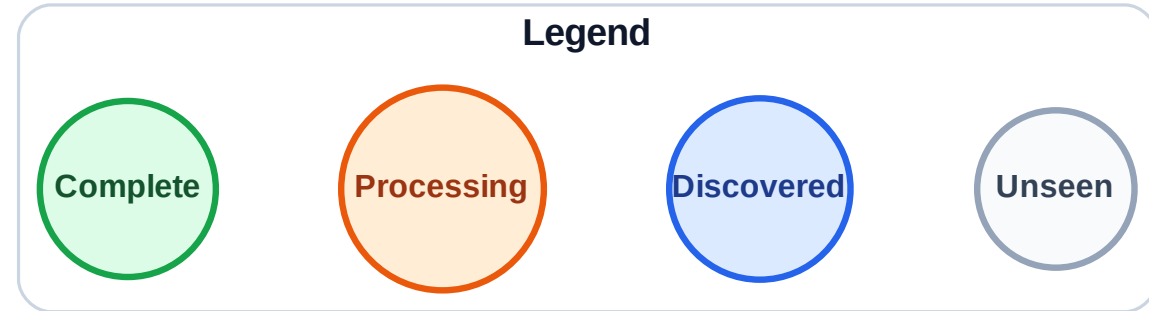
Queue: (empty)

Visited: (none)

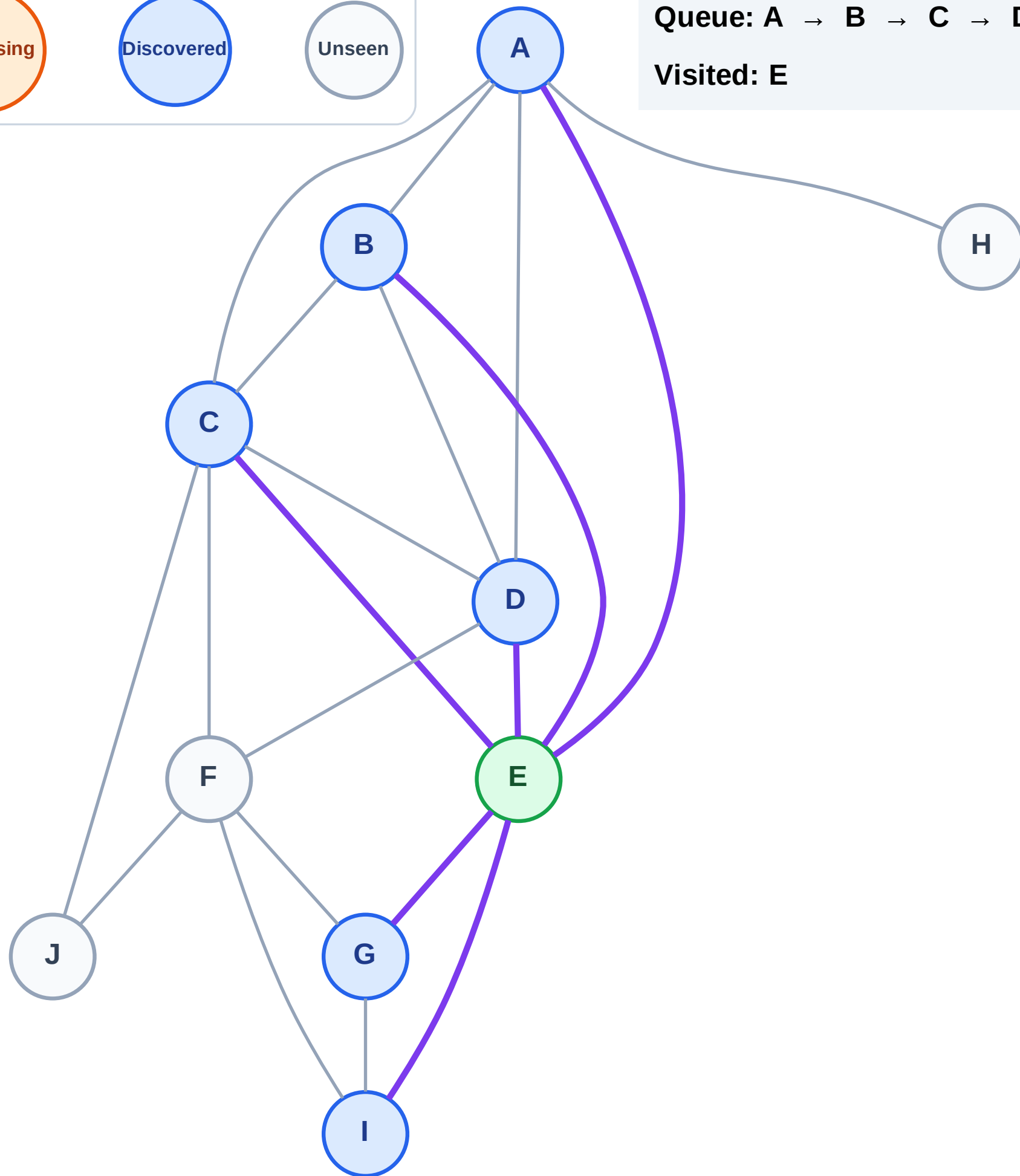


BFS Traversal

Step 3: Discover A, B, C, D, G, I from E



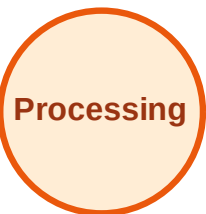
Queue: A → B → C → D → G → I
Visited: E



BFS Traversal

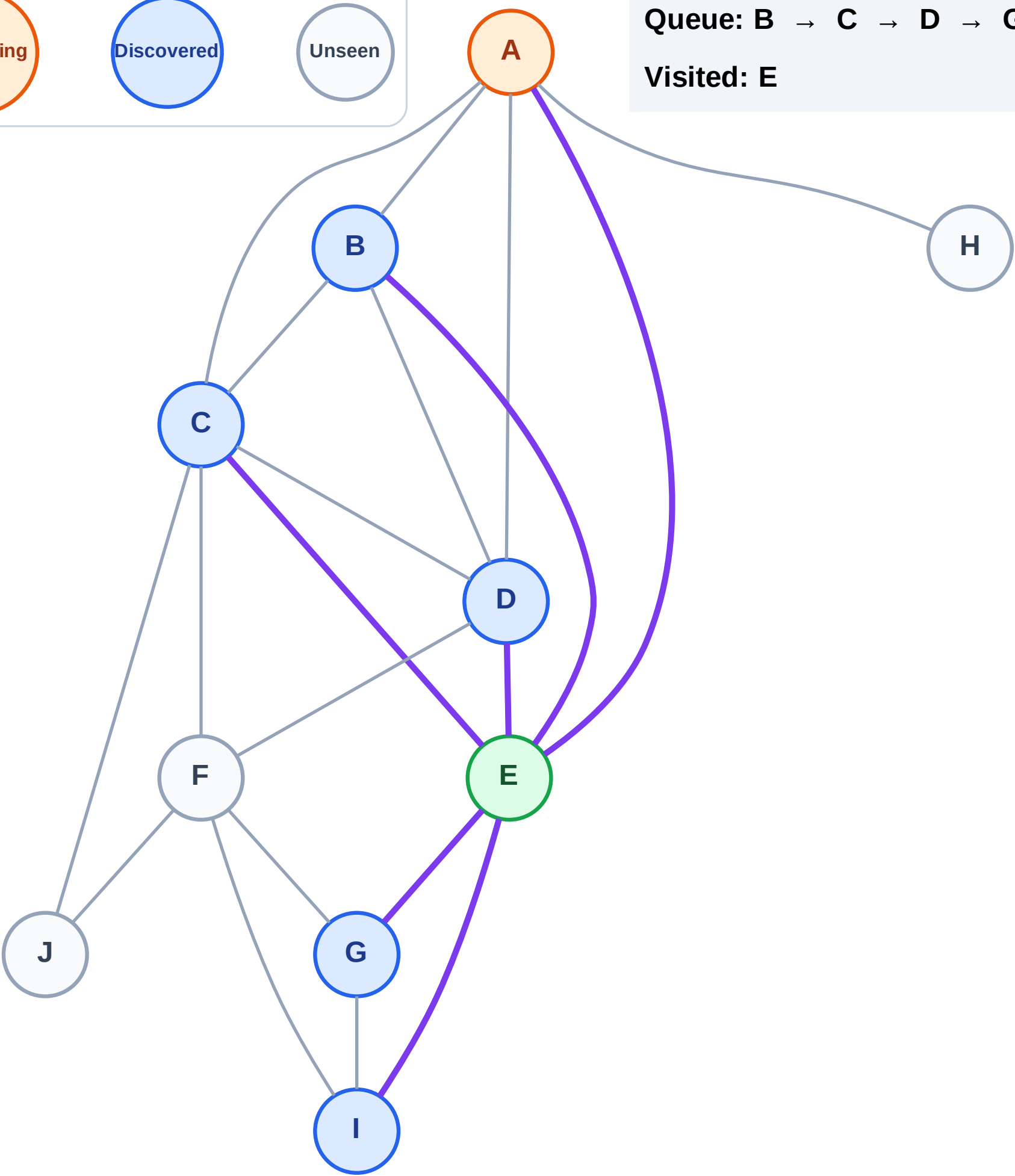
Step 4: Process node A

Legend



Queue: B → C → D → G → I

Visited: E



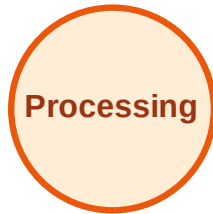
BFS Traversal

Step 5: Discover H from A

Legend



Complete



Processing

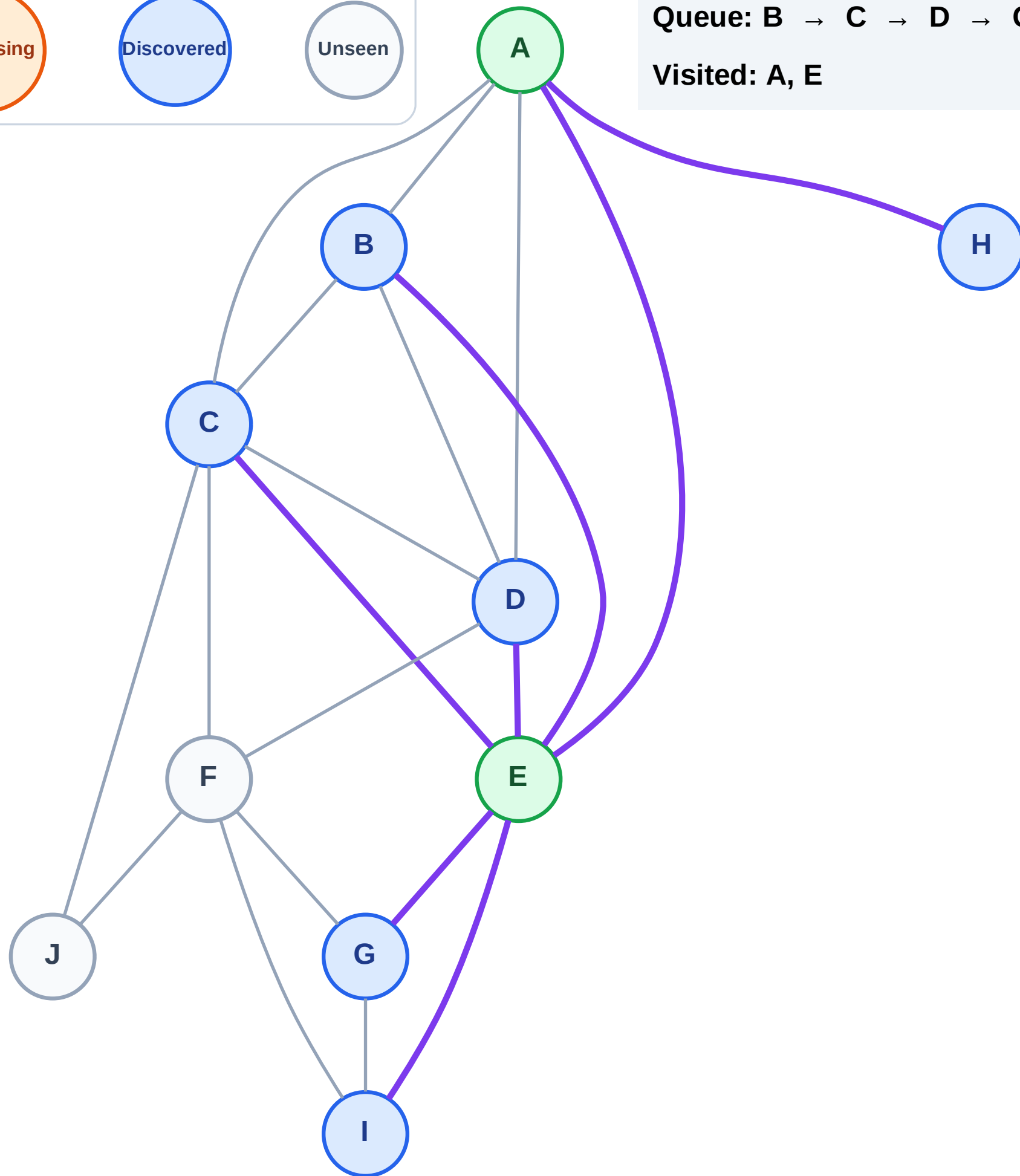


Discovered



Unseen

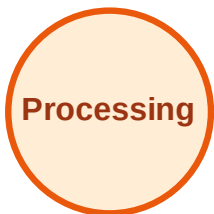
Queue: B → C → D → G → I → H
Visited: A, E



BFS Traversal

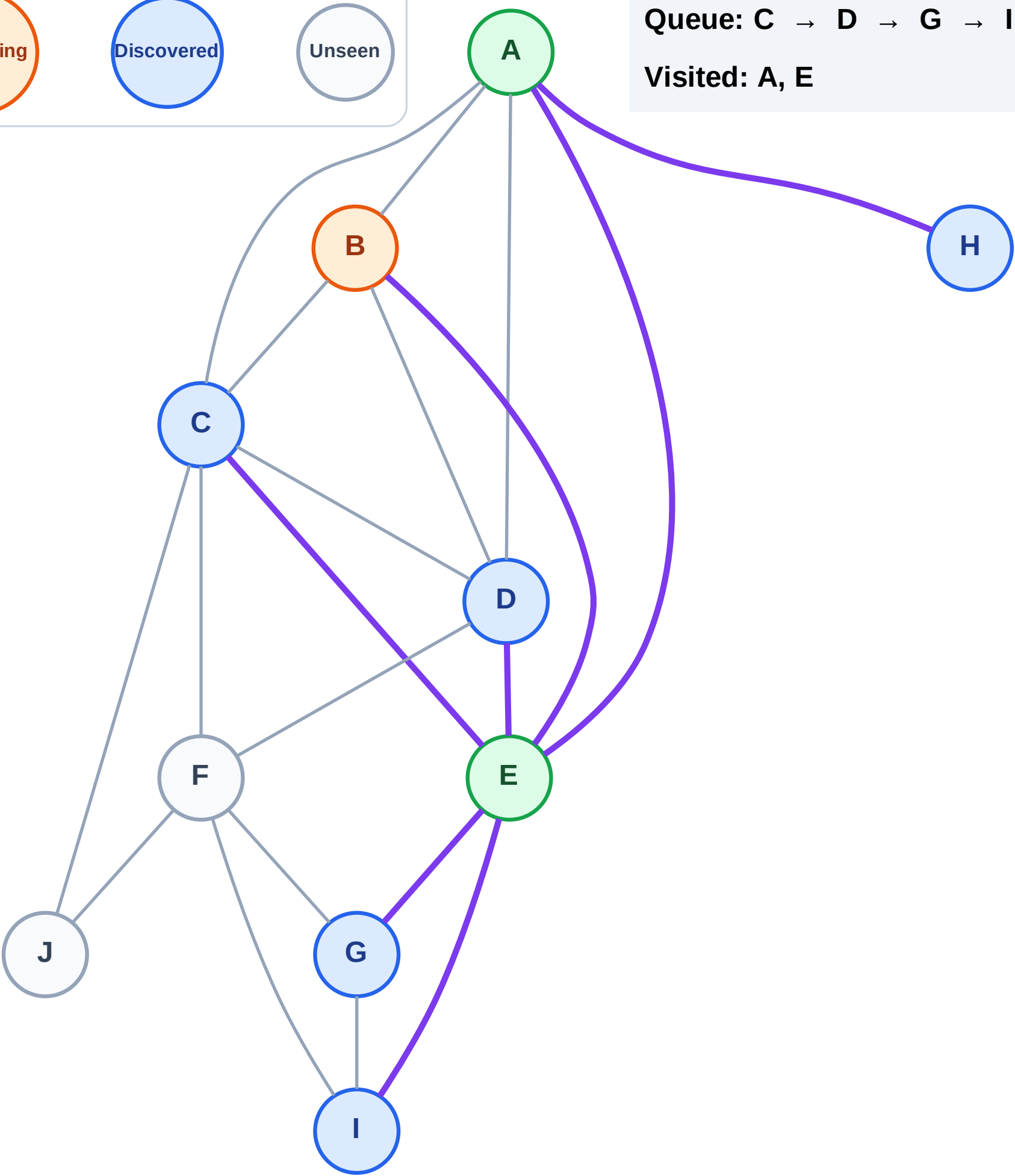
Step 6: Process node B

Legend



Queue: C → D → G → I → H

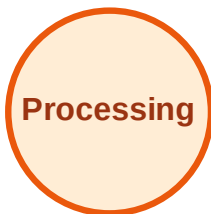
Visited: A, E



BFS Traversal

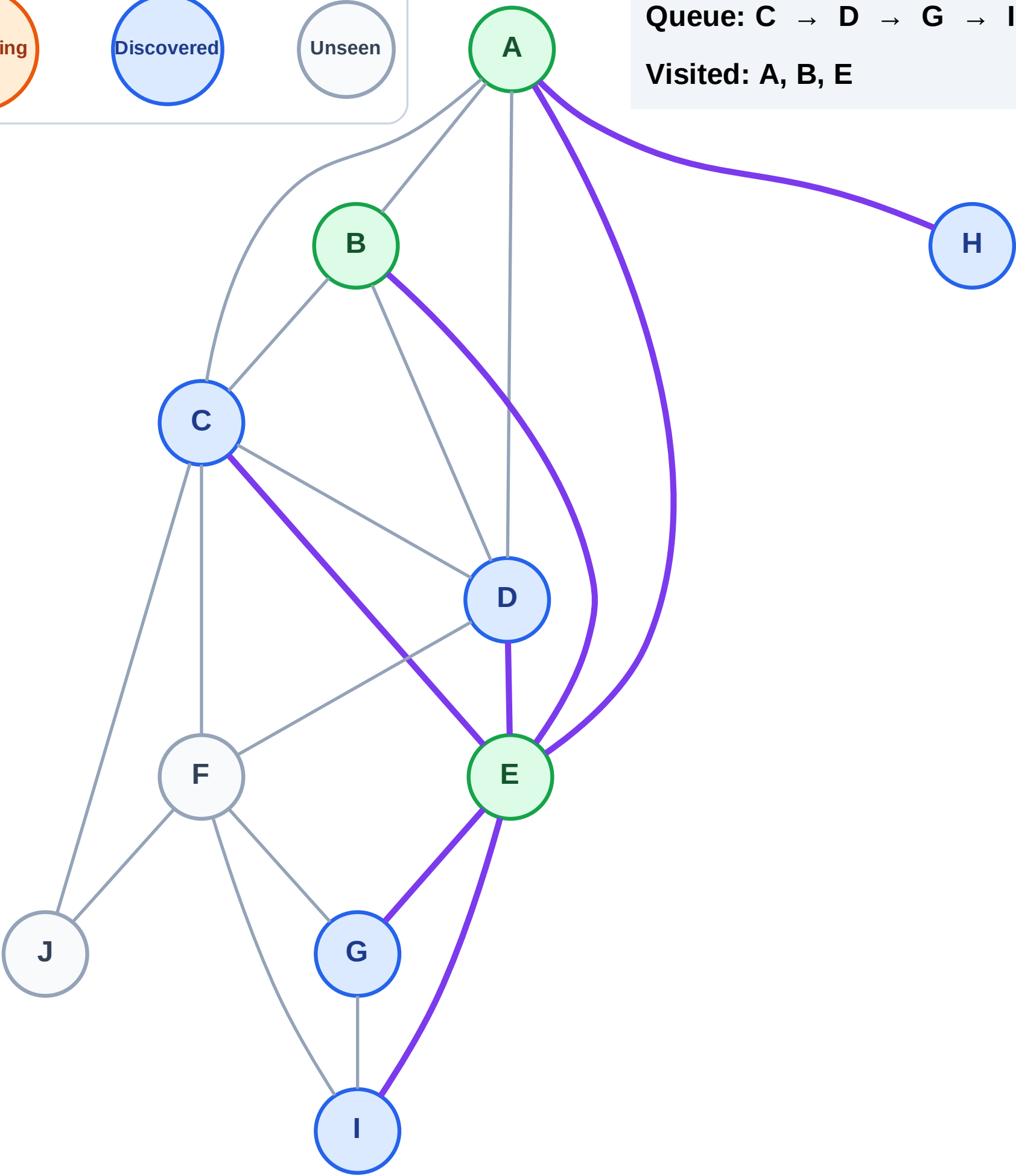
Step 7: No new neighbors from B

Legend



Queue: C → D → G → I → H

Visited: A, B, E



BFS Traversal

Step 8: Process node C

Legend

Complete

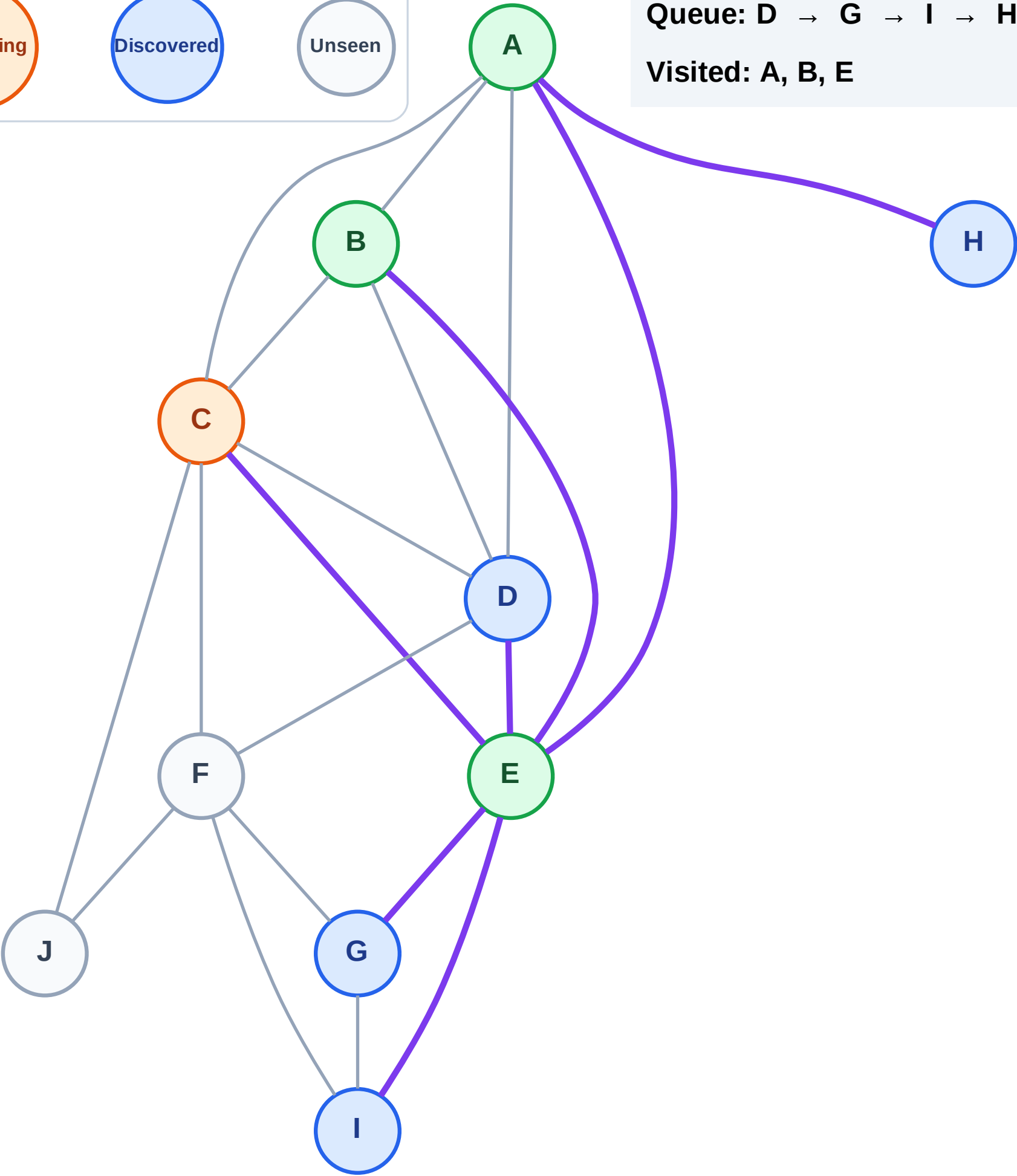
Processing

Discovered

Unseen

Queue: D → G → I → H

Visited: A, B, E



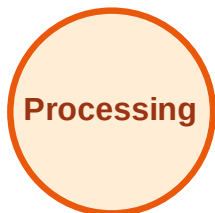
BFS Traversal

Step 9: Discover F, J from C

Legend



Complete



Processing



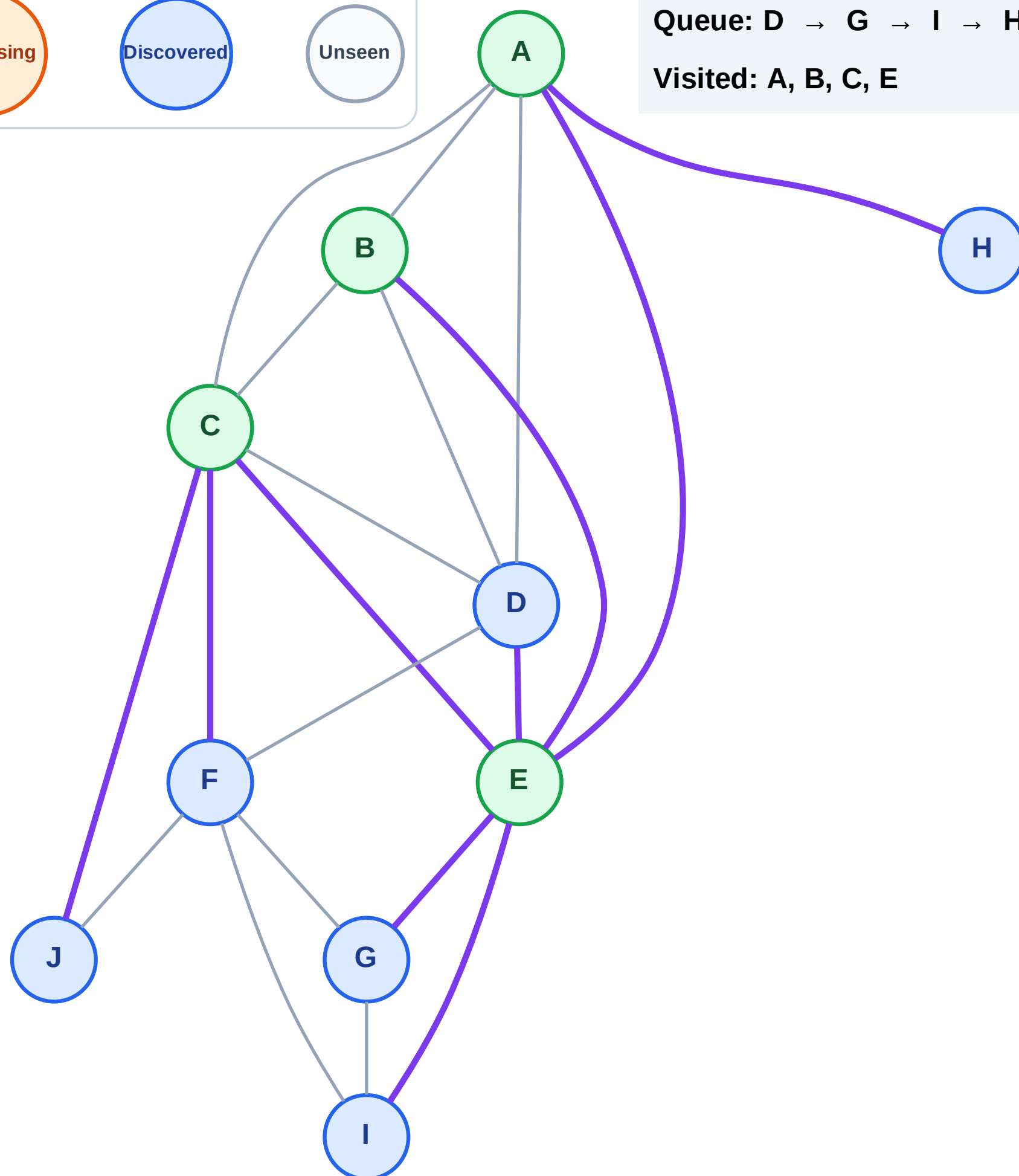
Discovered



Unseen

Queue: D → G → I → H → F → J

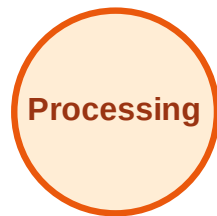
Visited: A, B, C, E



BFS Traversal

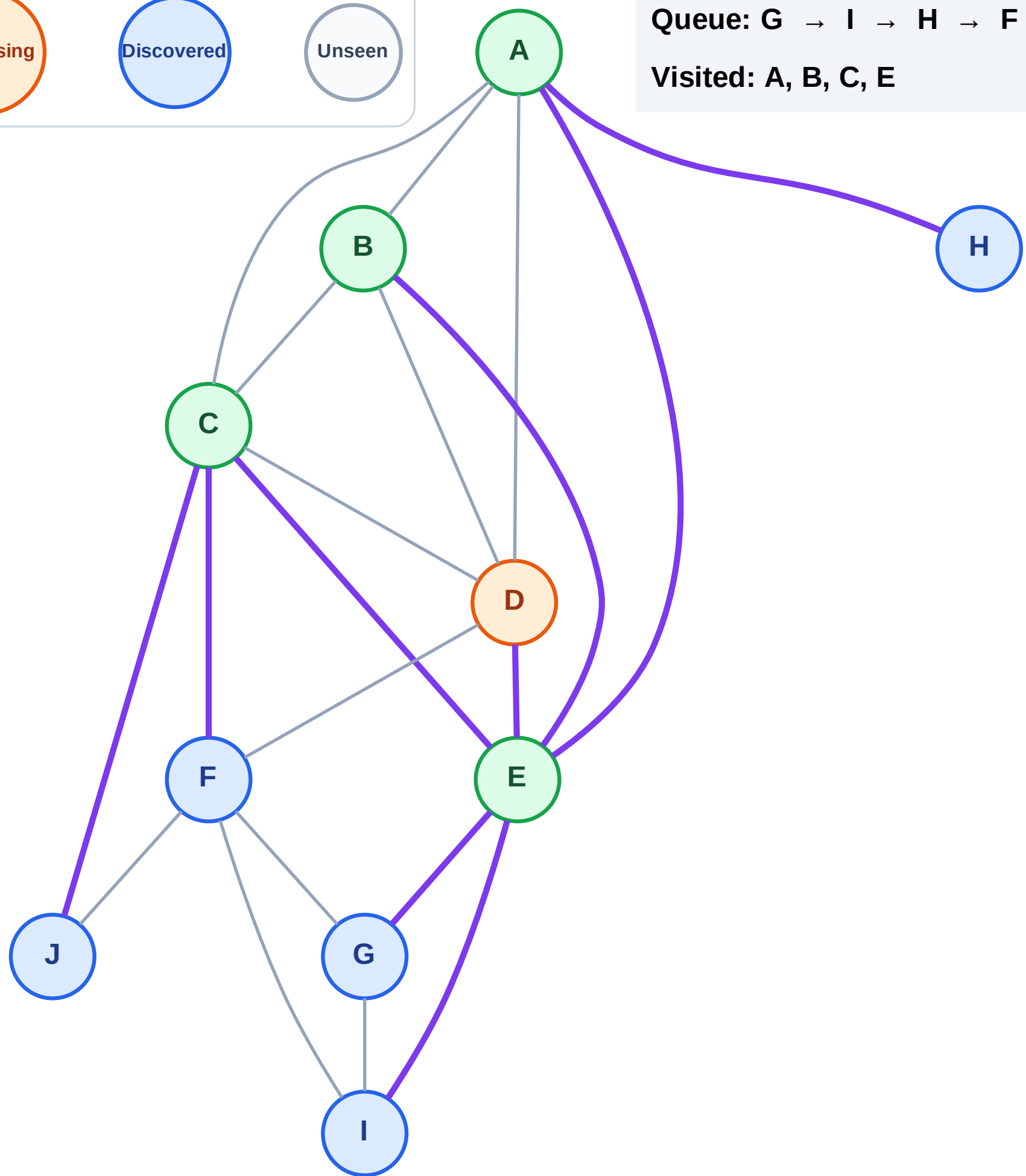
Step 10: Process node D

Legend



Queue: G → I → H → F → J

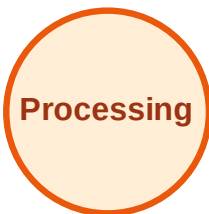
Visited: A, B, C, E



BFS Traversal

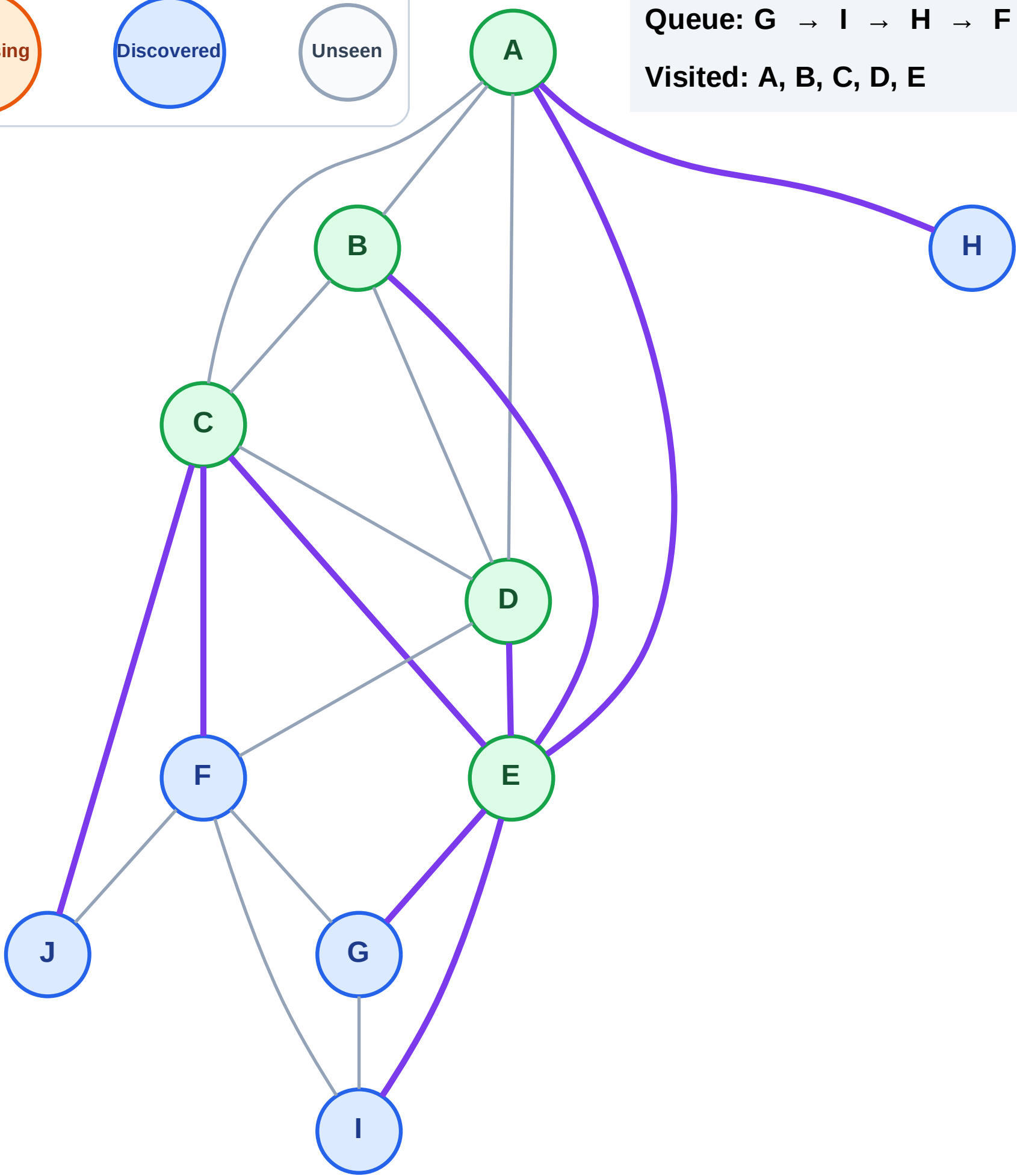
Step 11: No new neighbors from D

Legend



Queue: G → I → H → F → J

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node G

Legend

Complete

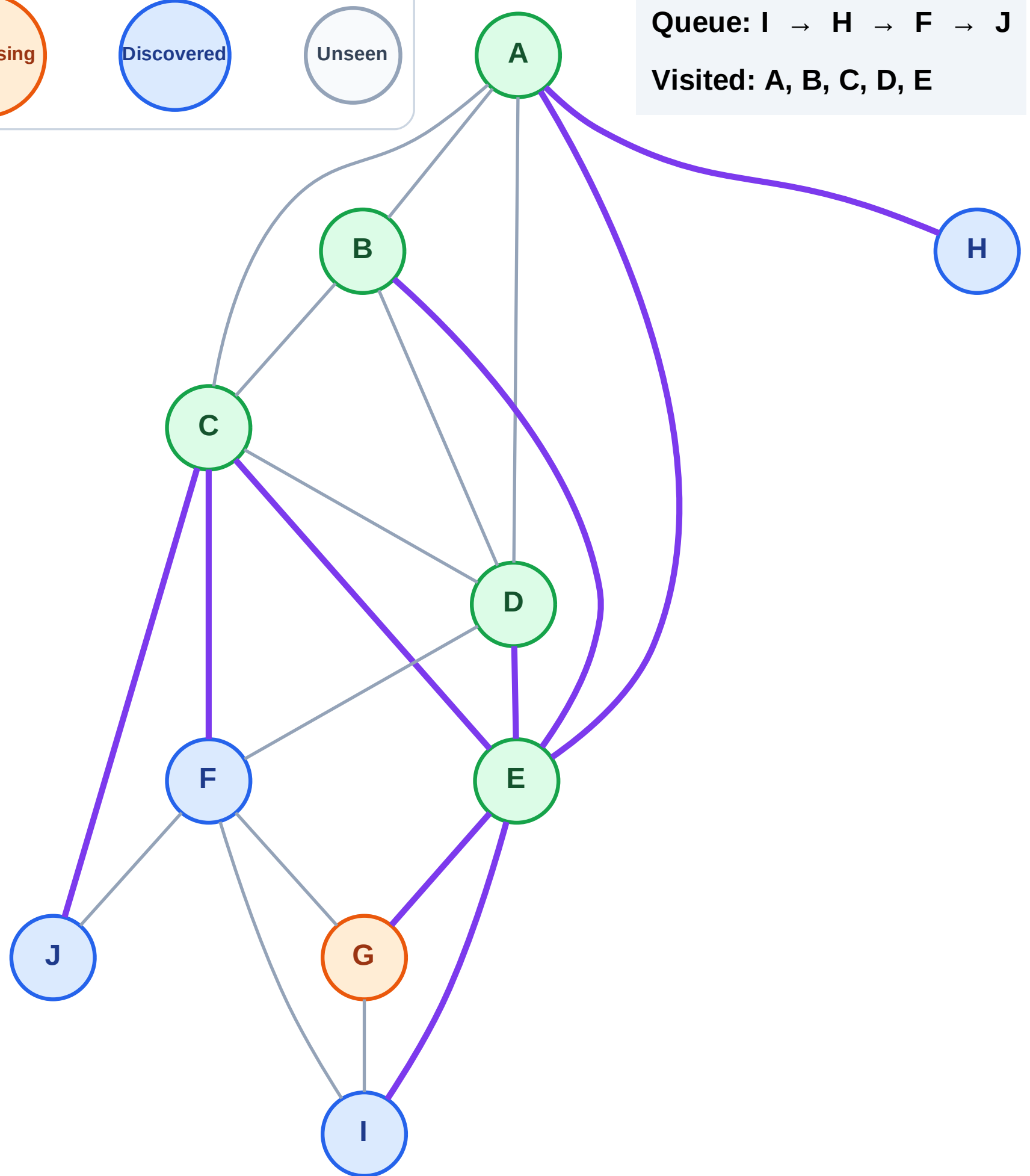
Processing

Discovered

Unseen

Queue: I → H → F → J

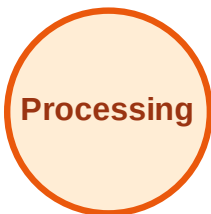
Visited: A, B, C, D, E



BFS Traversal

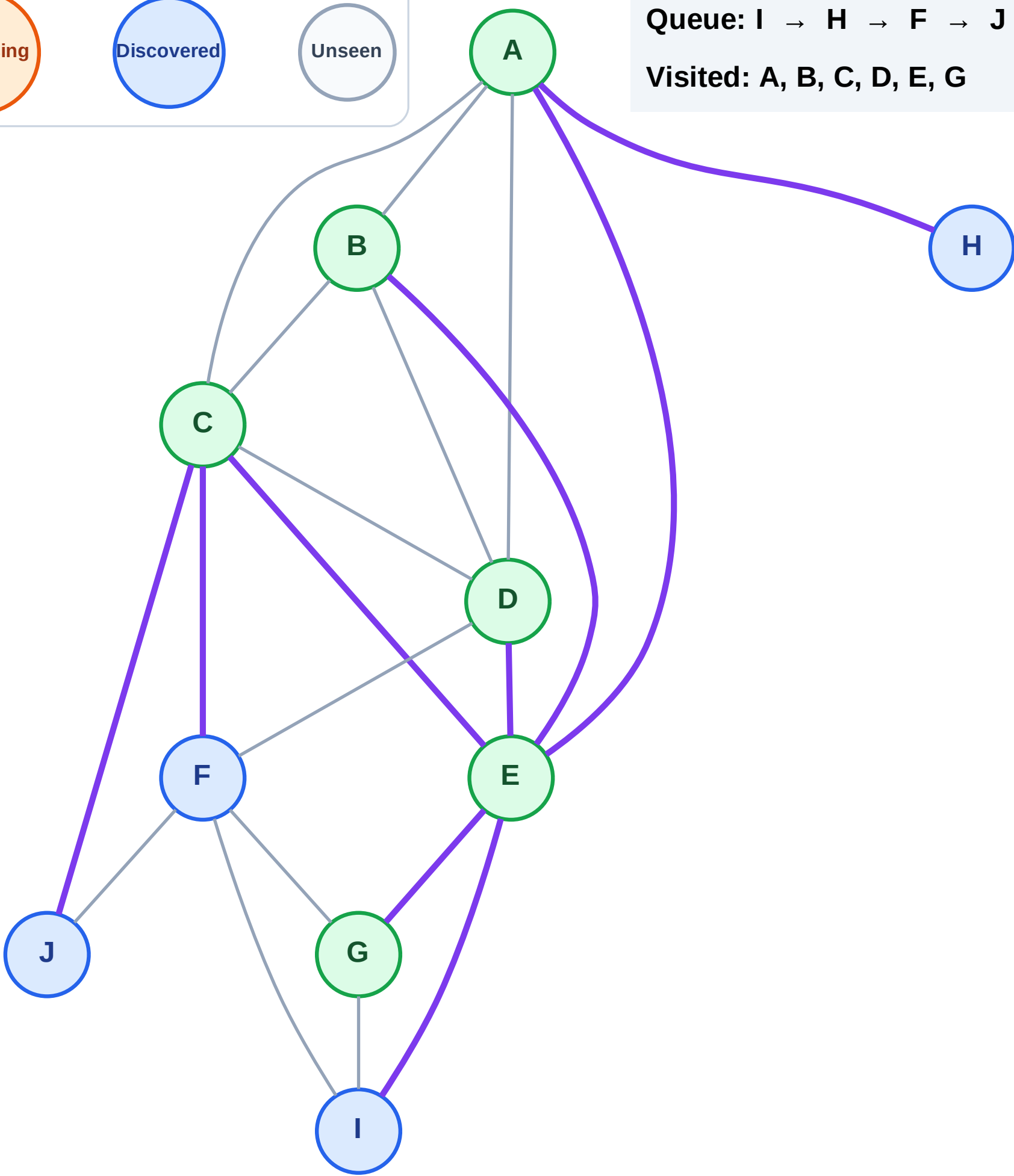
Step 13: No new neighbors from G

Legend



Queue: I → H → F → J

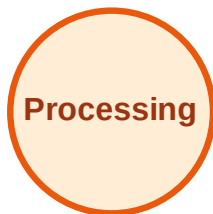
Visited: A, B, C, D, E, G



BFS Traversal

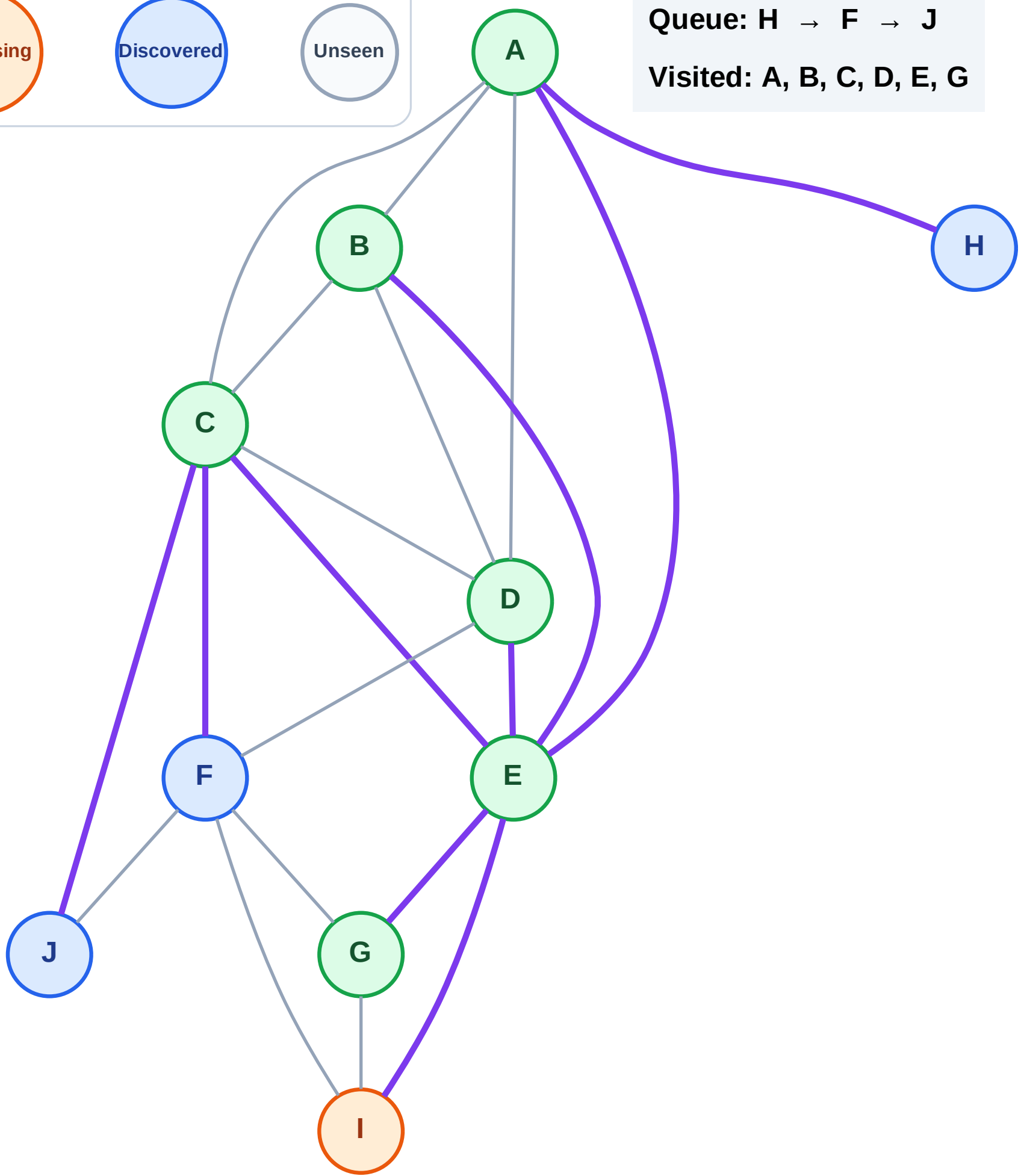
Step 14: Process node I

Legend



Queue: H → F → J

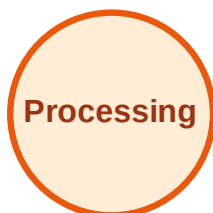
Visited: A, B, C, D, E, G



BFS Traversal

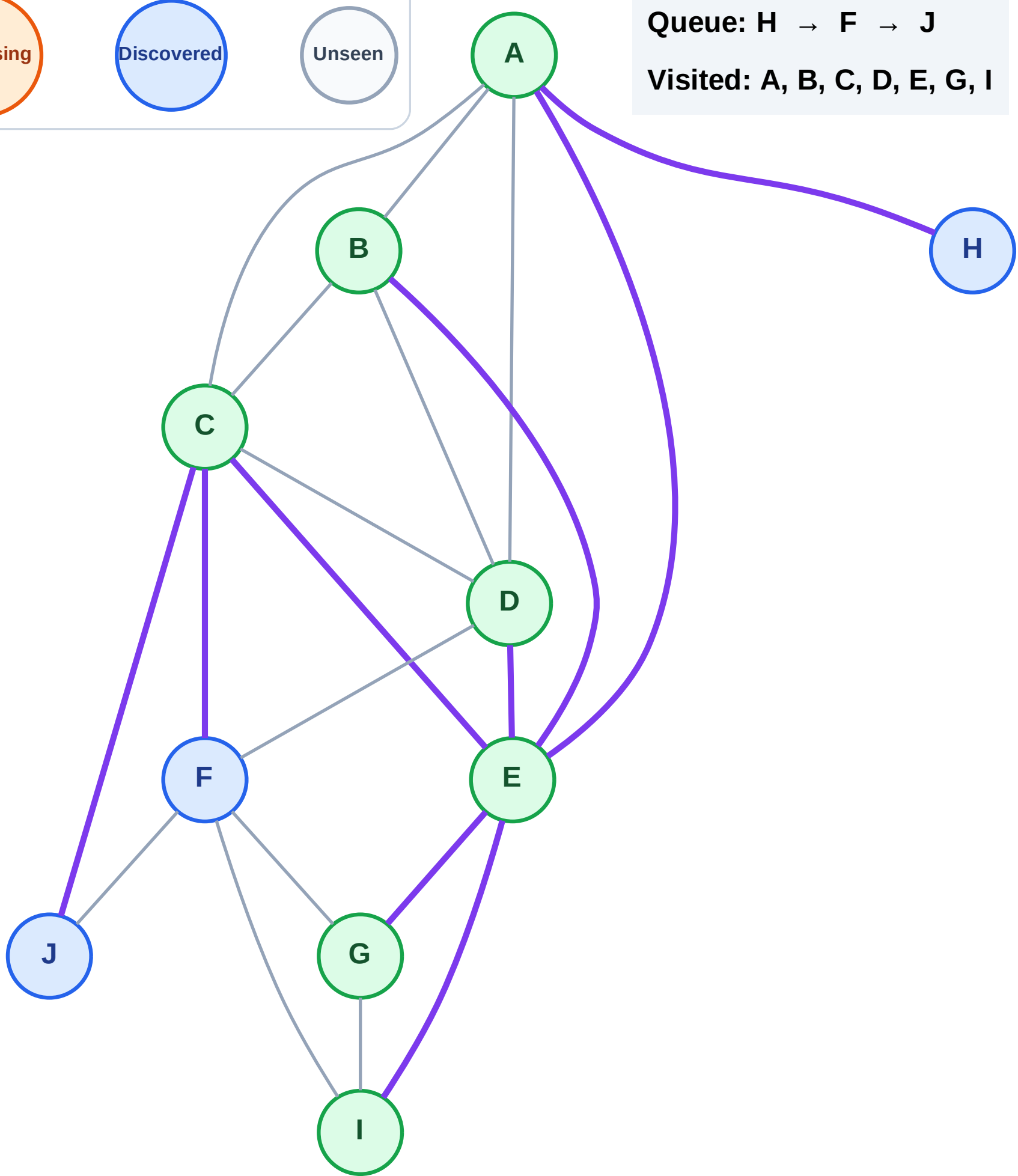
Step 15: No new neighbors from I

Legend



Queue: H → F → J

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node H

Legend

Complete

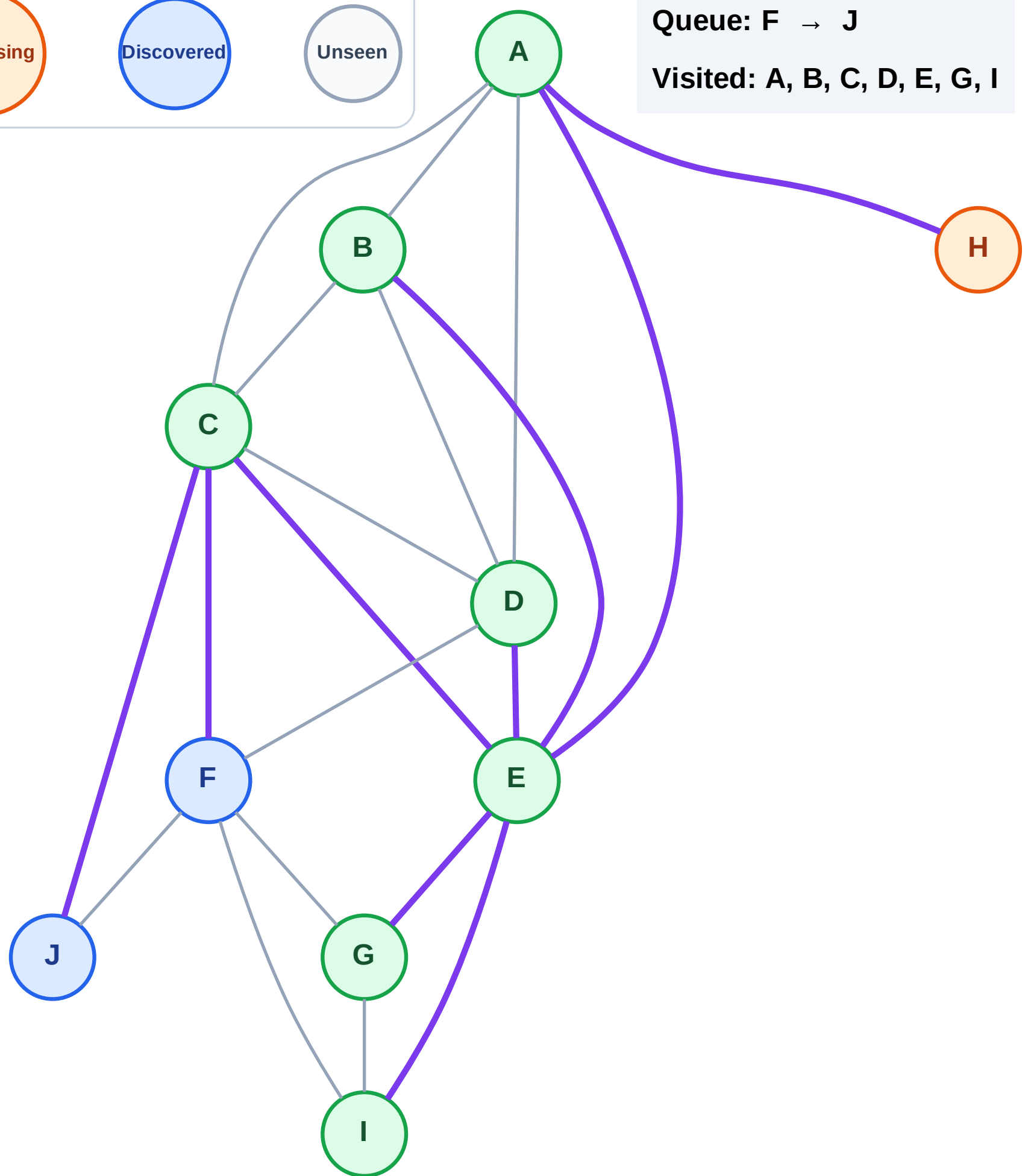
Processing

Discovered

Unseen

Queue: F → J

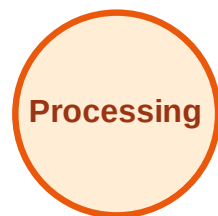
Visited: A, B, C, D, E, G, I



BFS Traversal

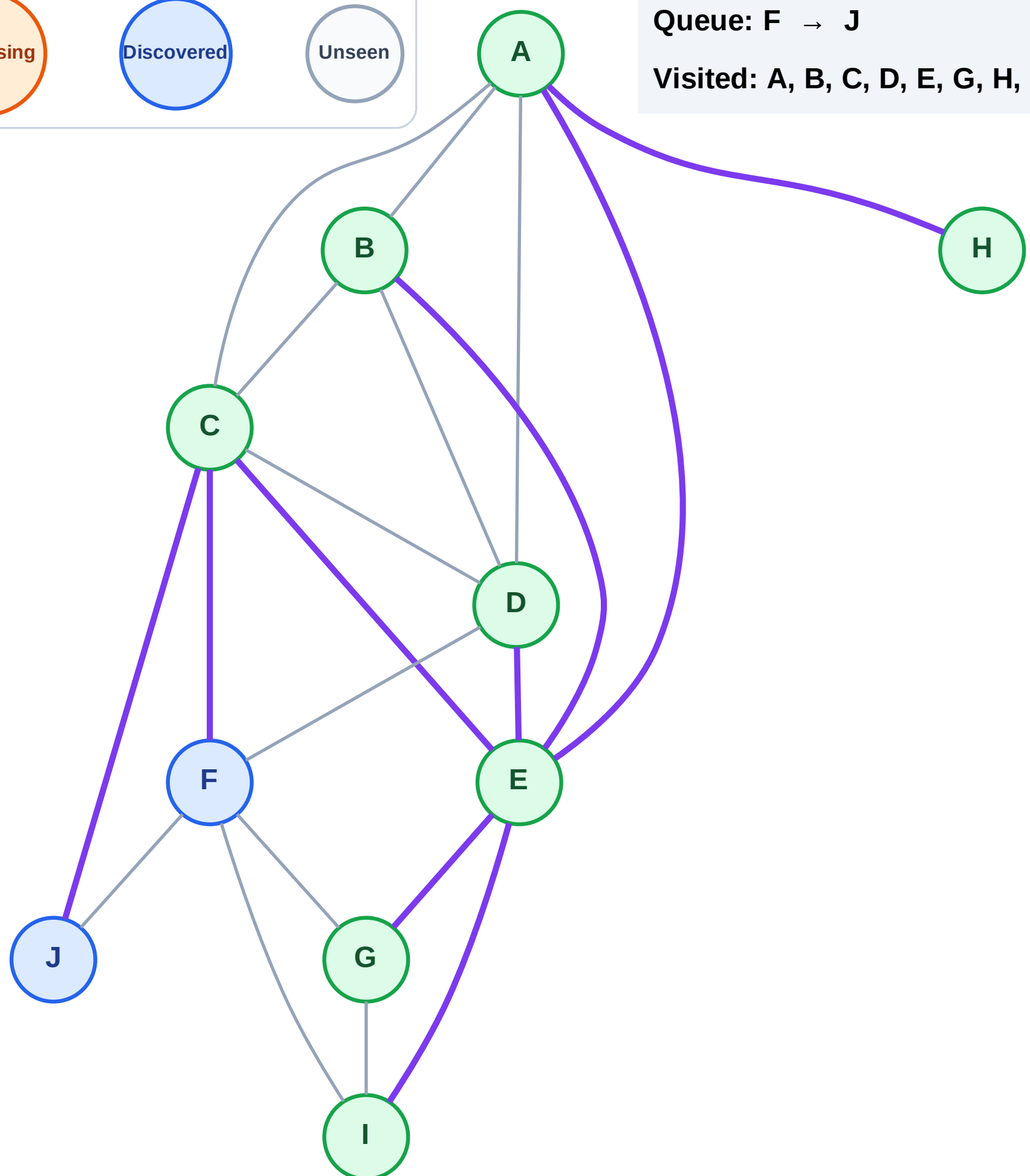
Step 17: No new neighbors from H

Legend



Queue: F → J

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 18: Process node F

Legend

Complete

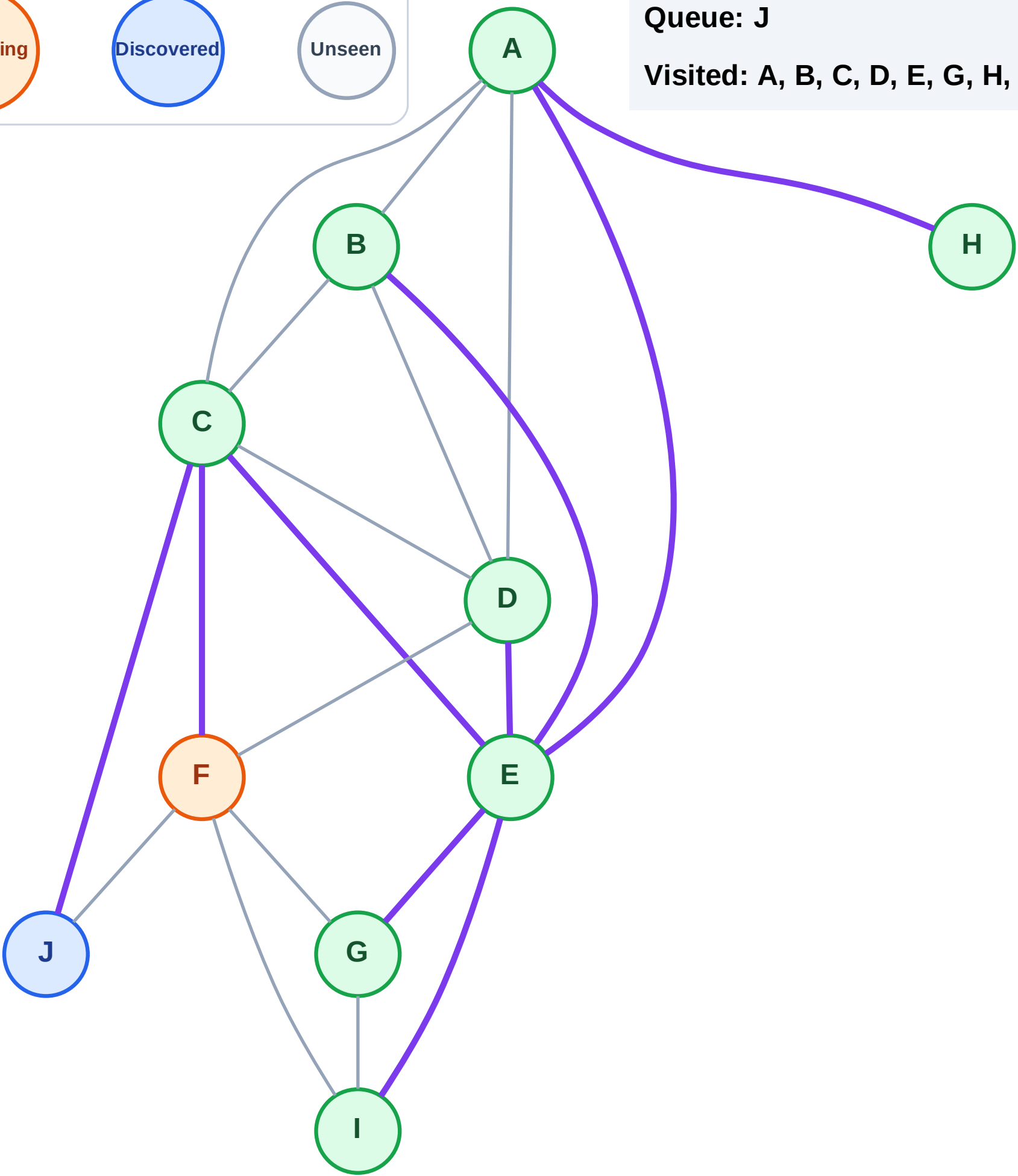
Processing

Discovered

Unseen

Queue: J

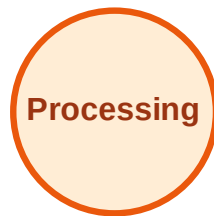
Visited: A, B, C, D, E, G, H, I



BFS Traversal

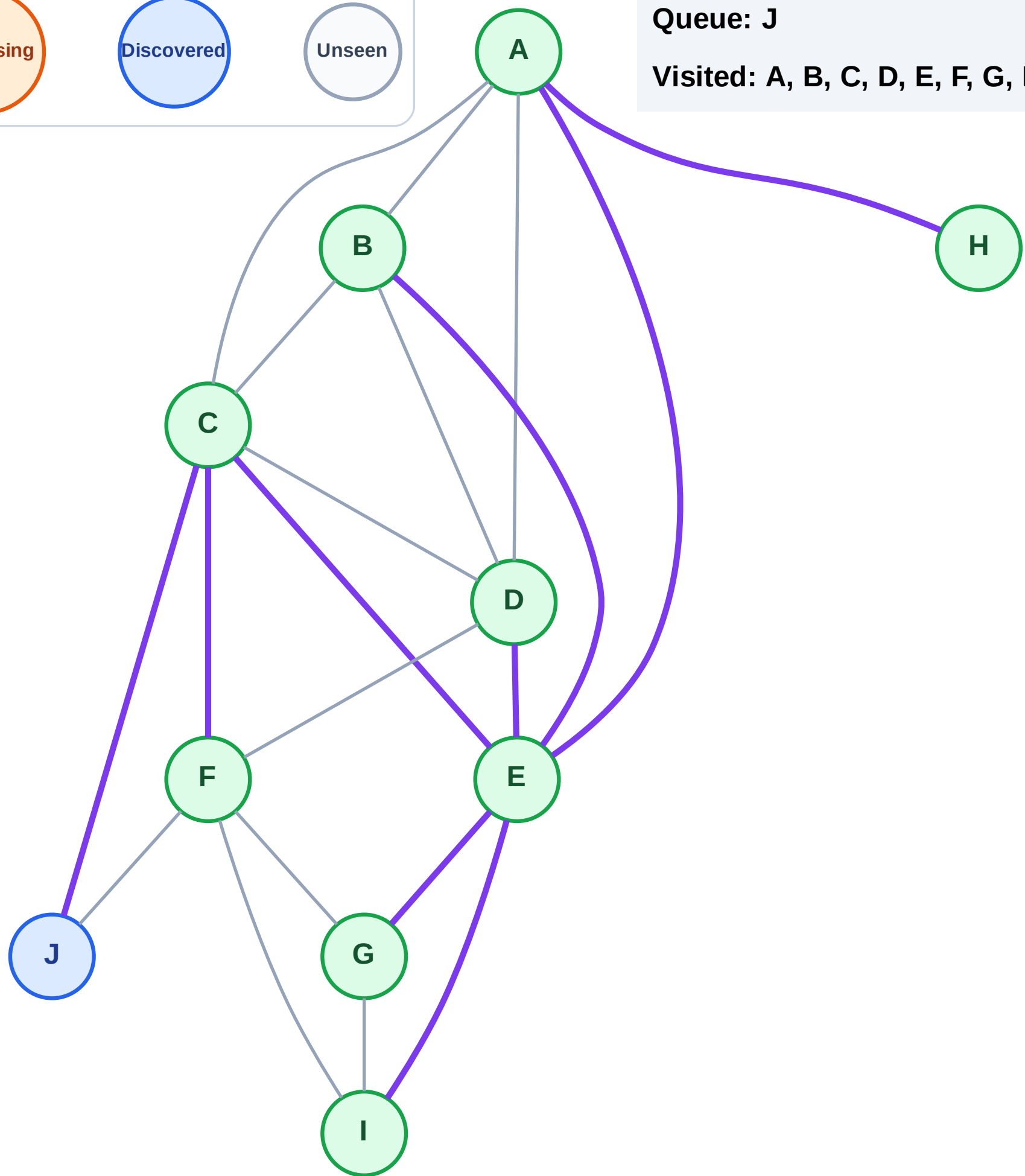
Step 19: No new neighbors from F

Legend



Queue: J

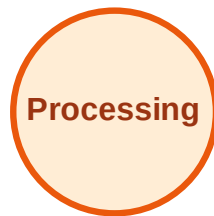
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

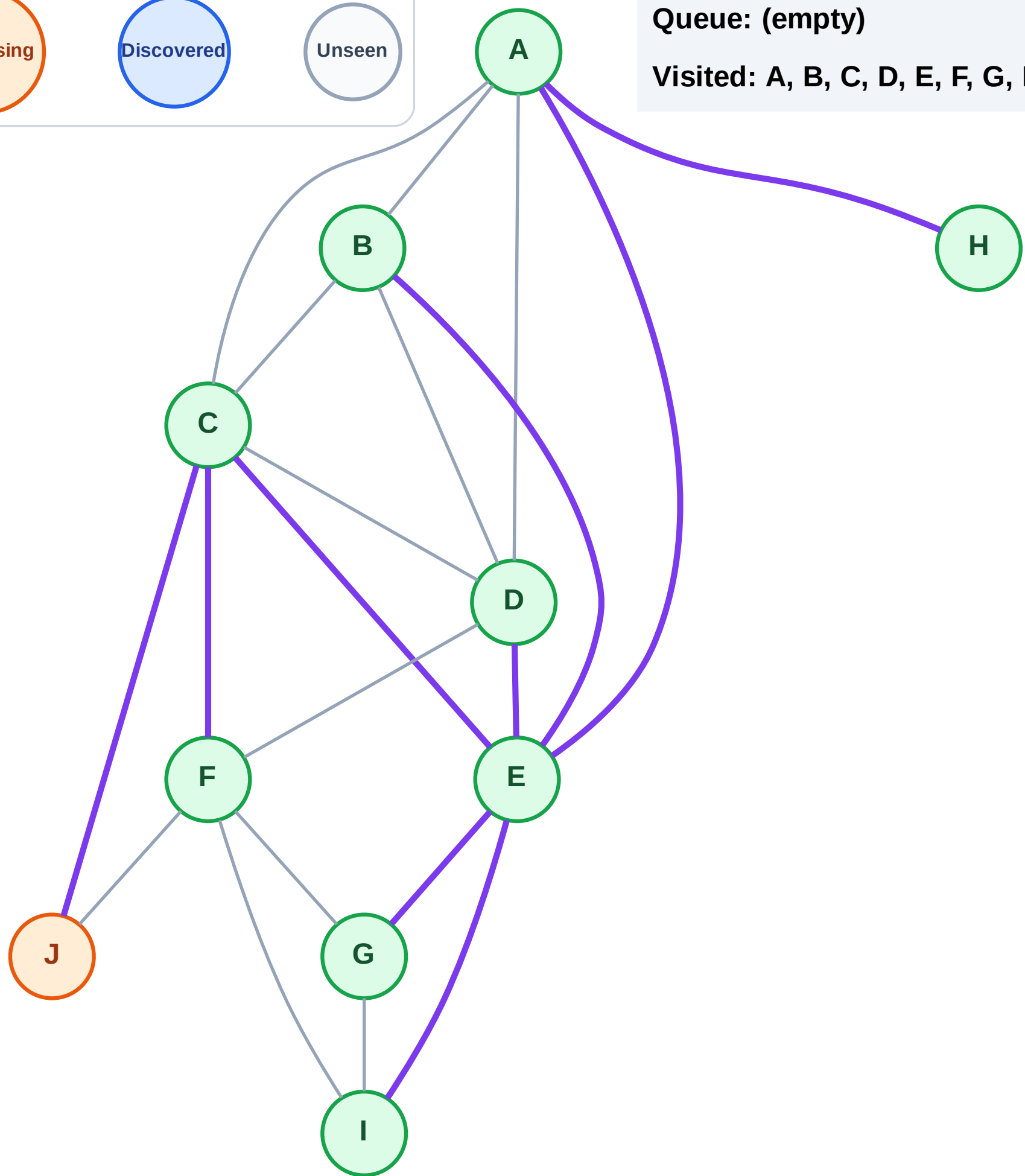
Step 20: Process node J

Legend



Queue: (empty)

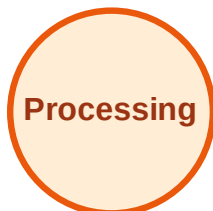
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

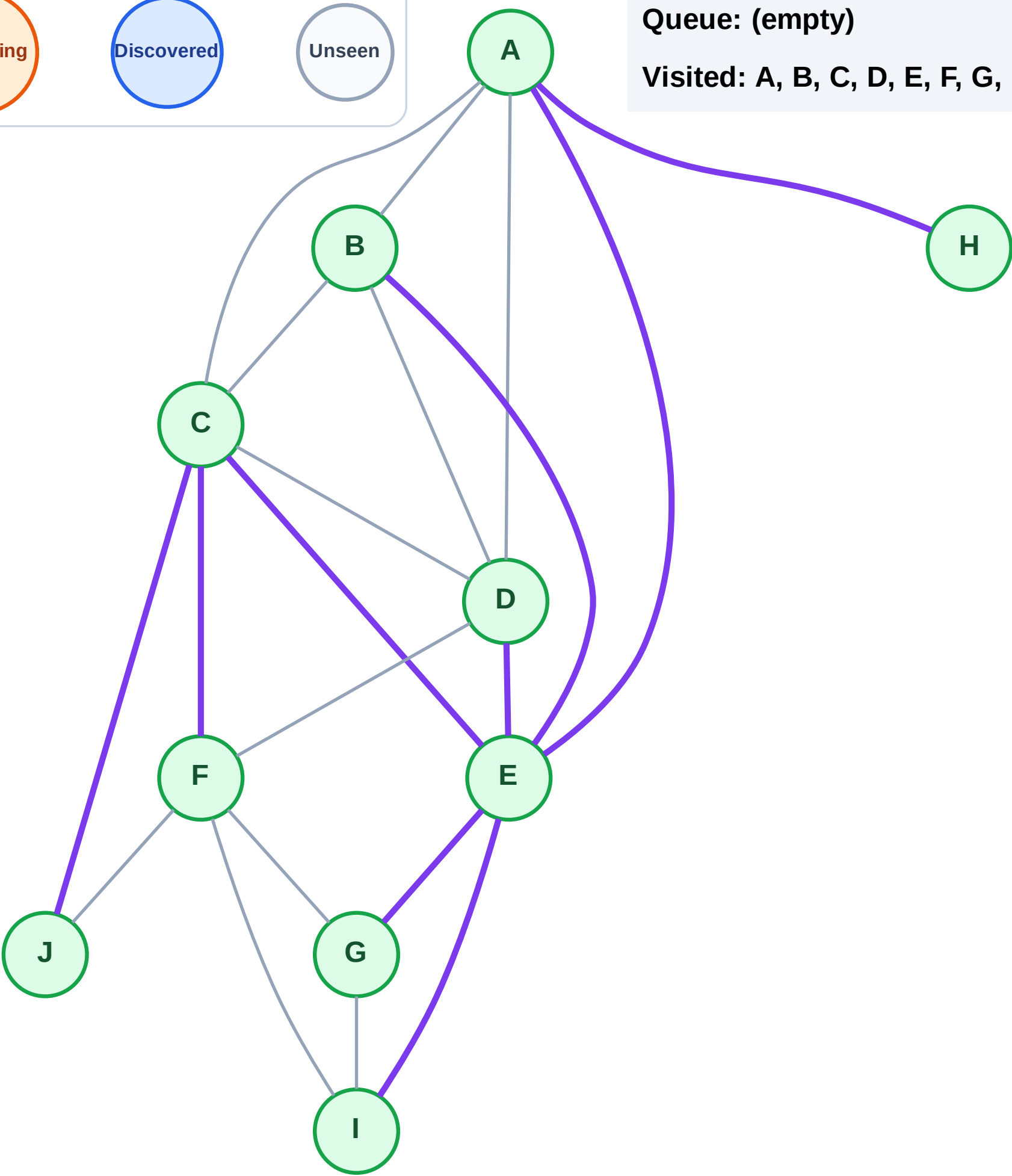
Step 21: No new neighbors from J

Legend



Queue: (empty)

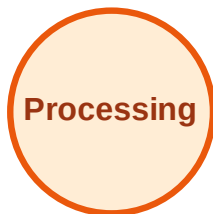
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

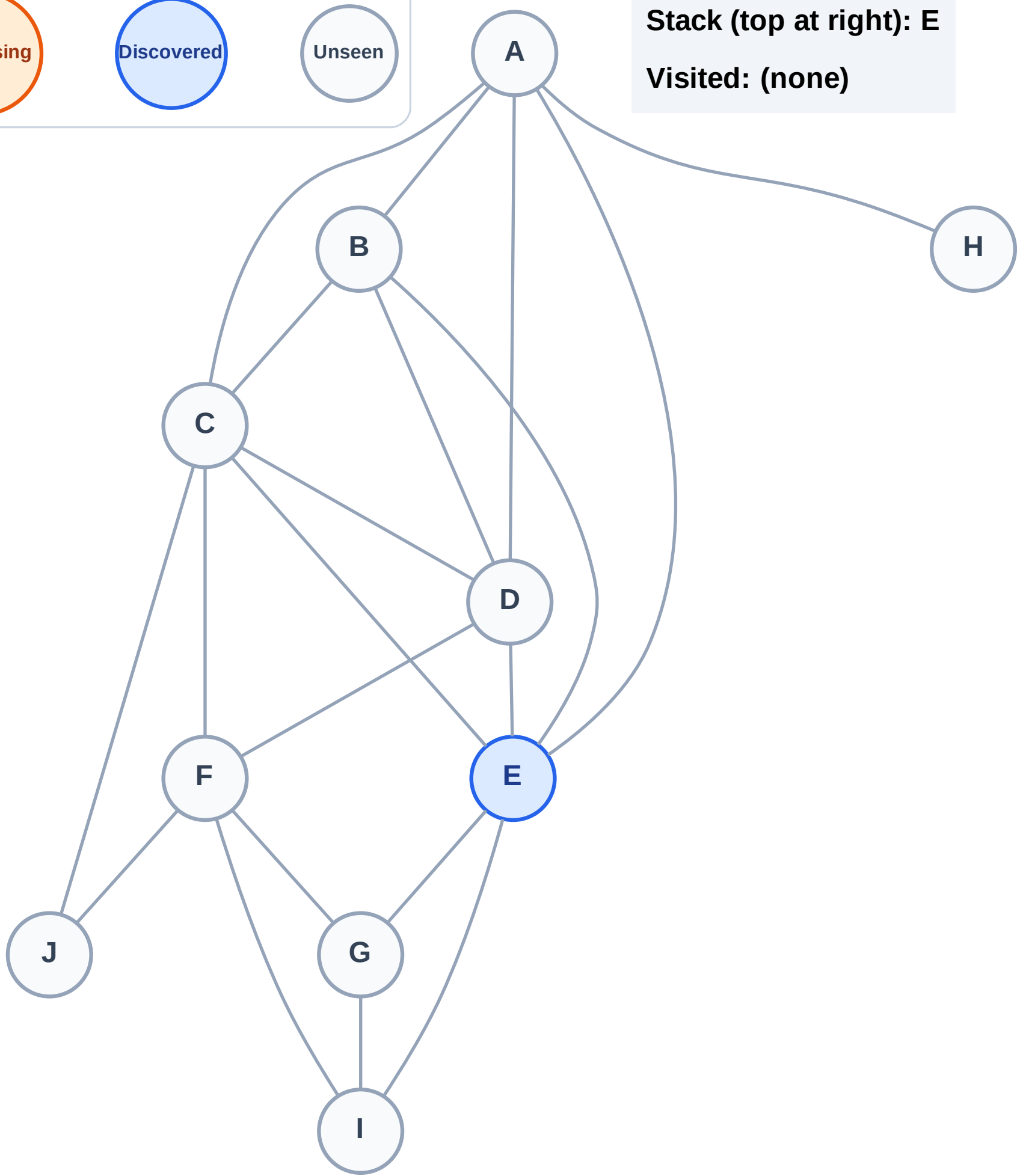
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

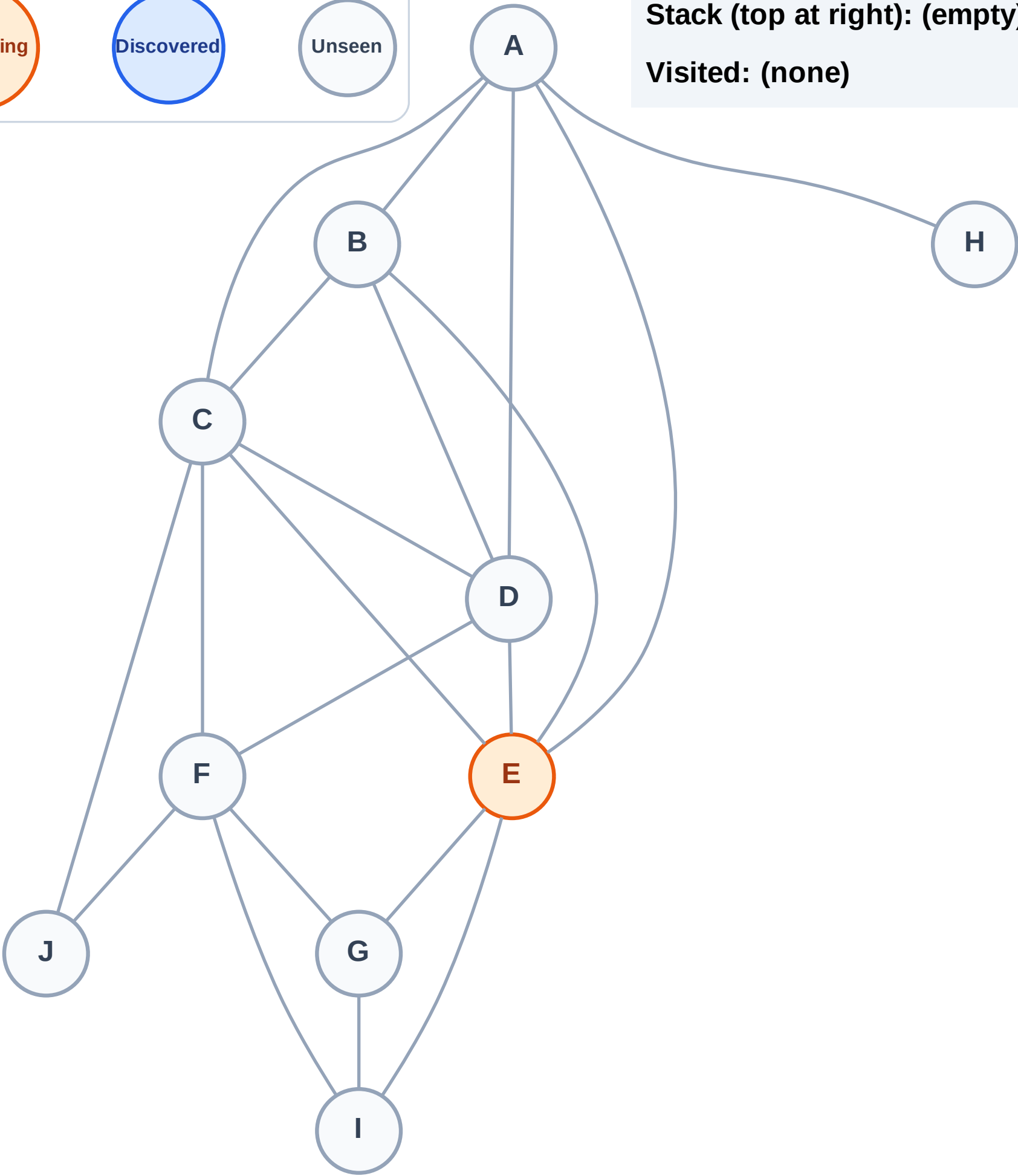
Complete

Processing

Discovered

Unseen

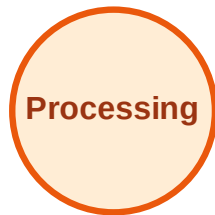
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

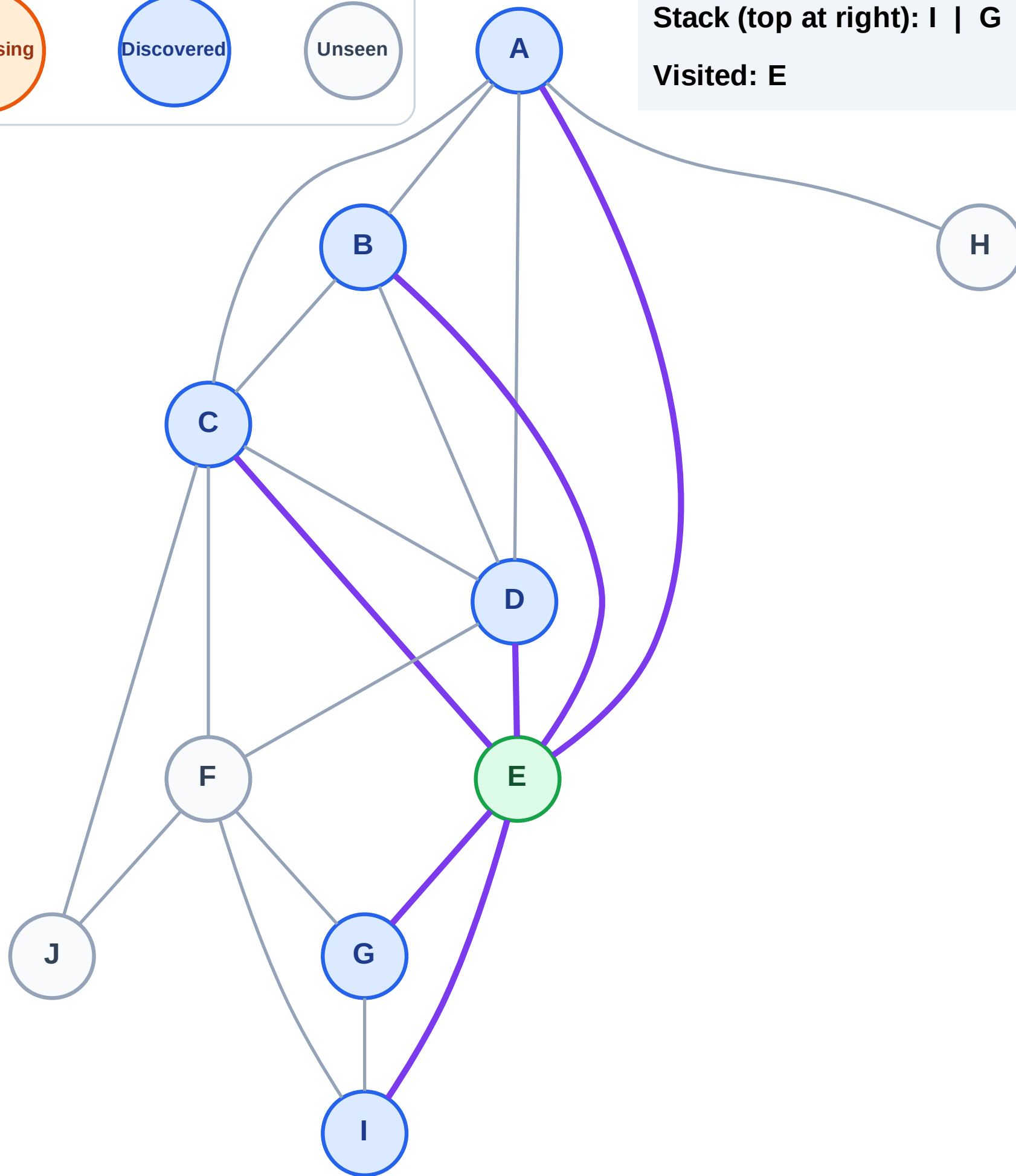
Step 3: Discover I, G, D, C, B, A from E

Legend



Stack (top at right): I | G | D | C | B | A

Visited: E



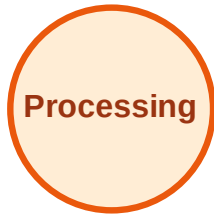
DFS Traversal

Step 4: Process node A

Legend



Complete



Processing



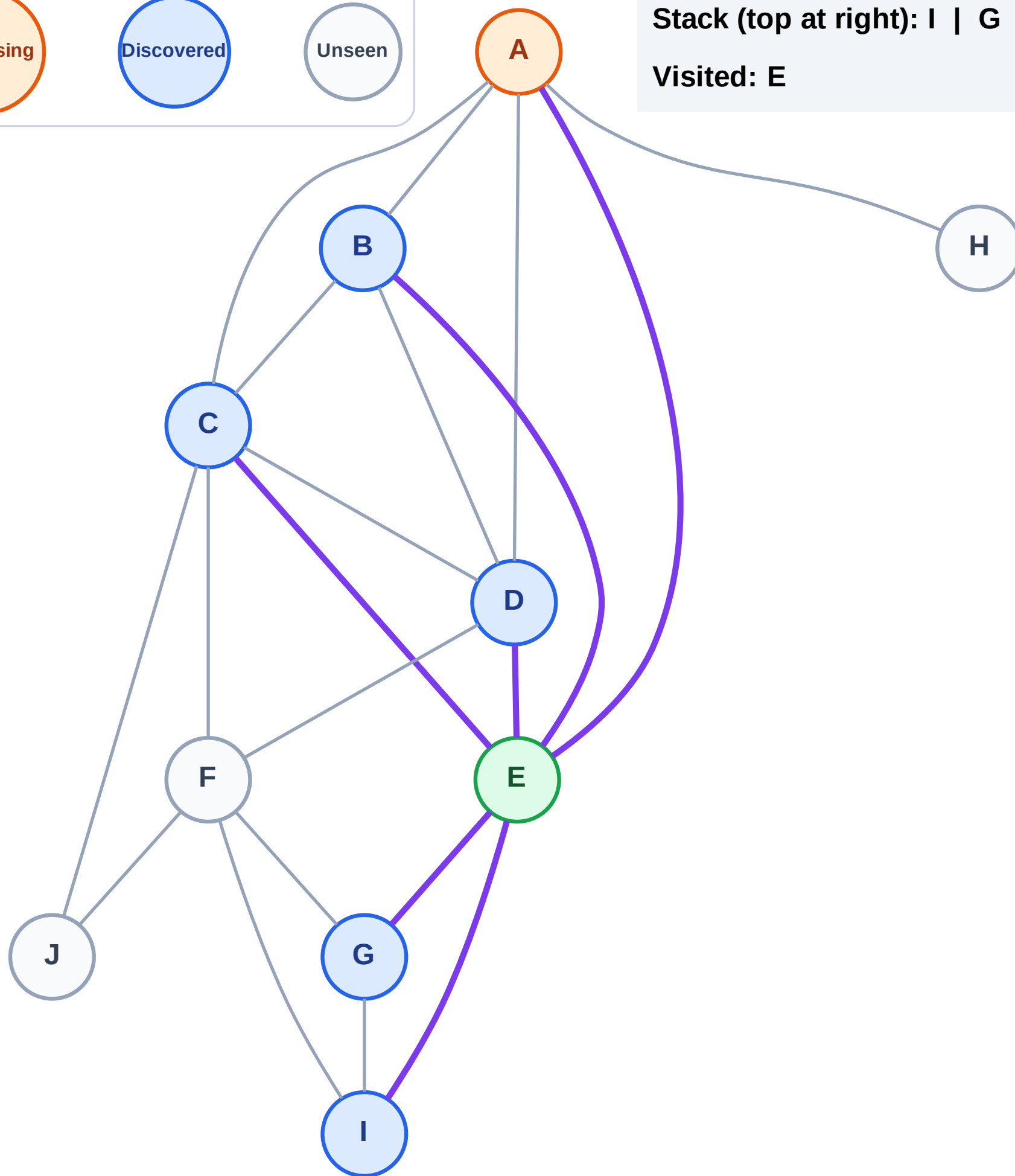
Discovered



Unseen

Stack (top at right): I | G | D | C | B

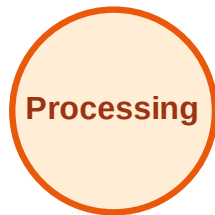
Visited: E



DFS Traversal

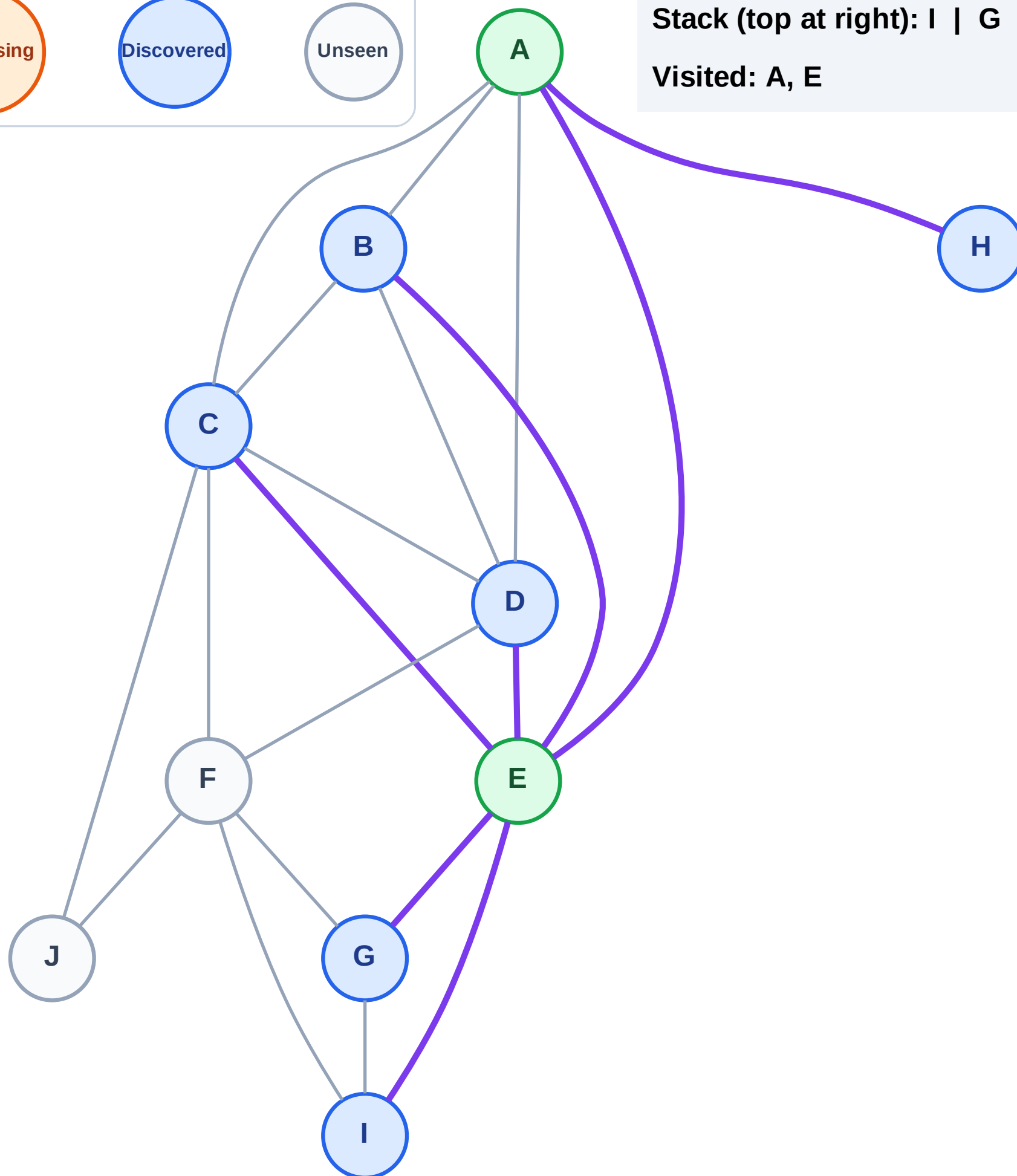
Step 5: Discover H from A

Legend



Stack (top at right): I | G | D | C | B | H

Visited: A, E



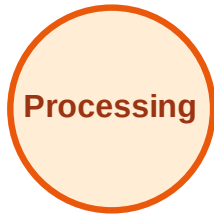
DFS Traversal

Step 6: Process node H

Legend



Complete



Processing



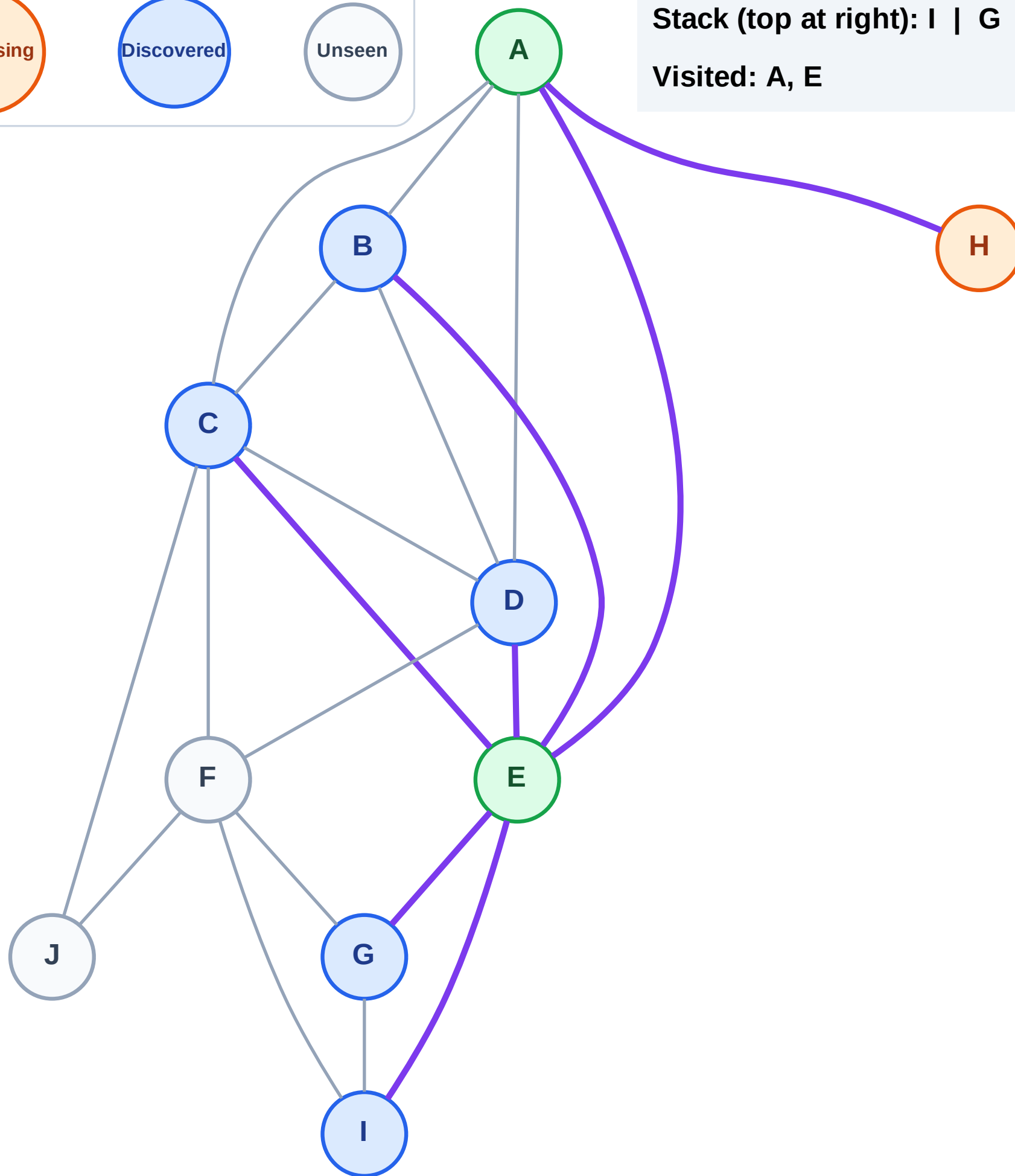
Discovered



Unseen

Stack (top at right): I | G | D | C | B

Visited: A, E

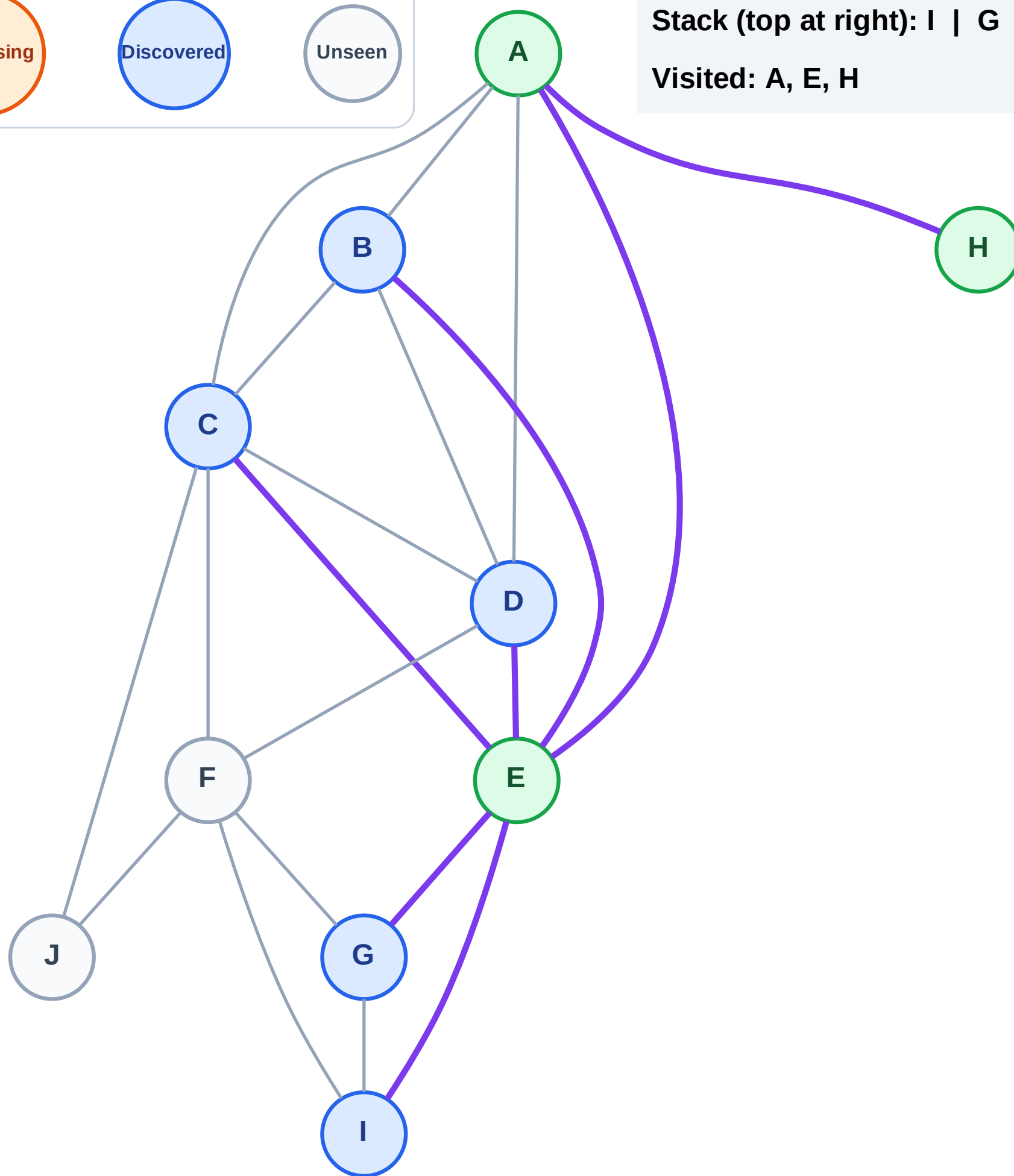


Step 7: No new neighbors from H

Legend

Unseen

Visited: A, E, H



DFS Traversal

Step 8: Process node B

Legend

Complete

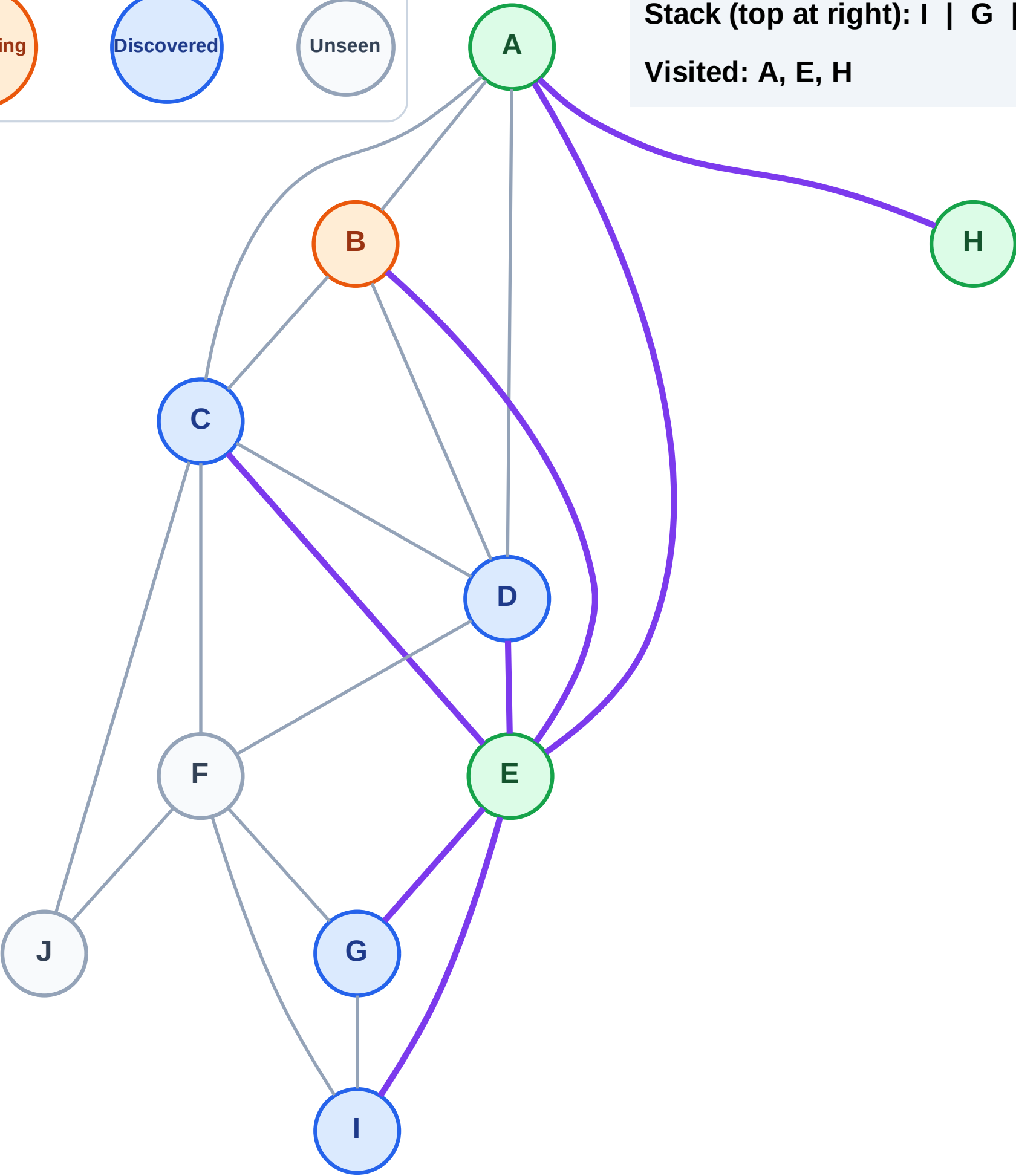
Processing

Discovered

Unseen

Stack (top at right): I | G | D | C

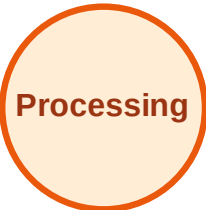
Visited: A, E, H



DFS Traversal

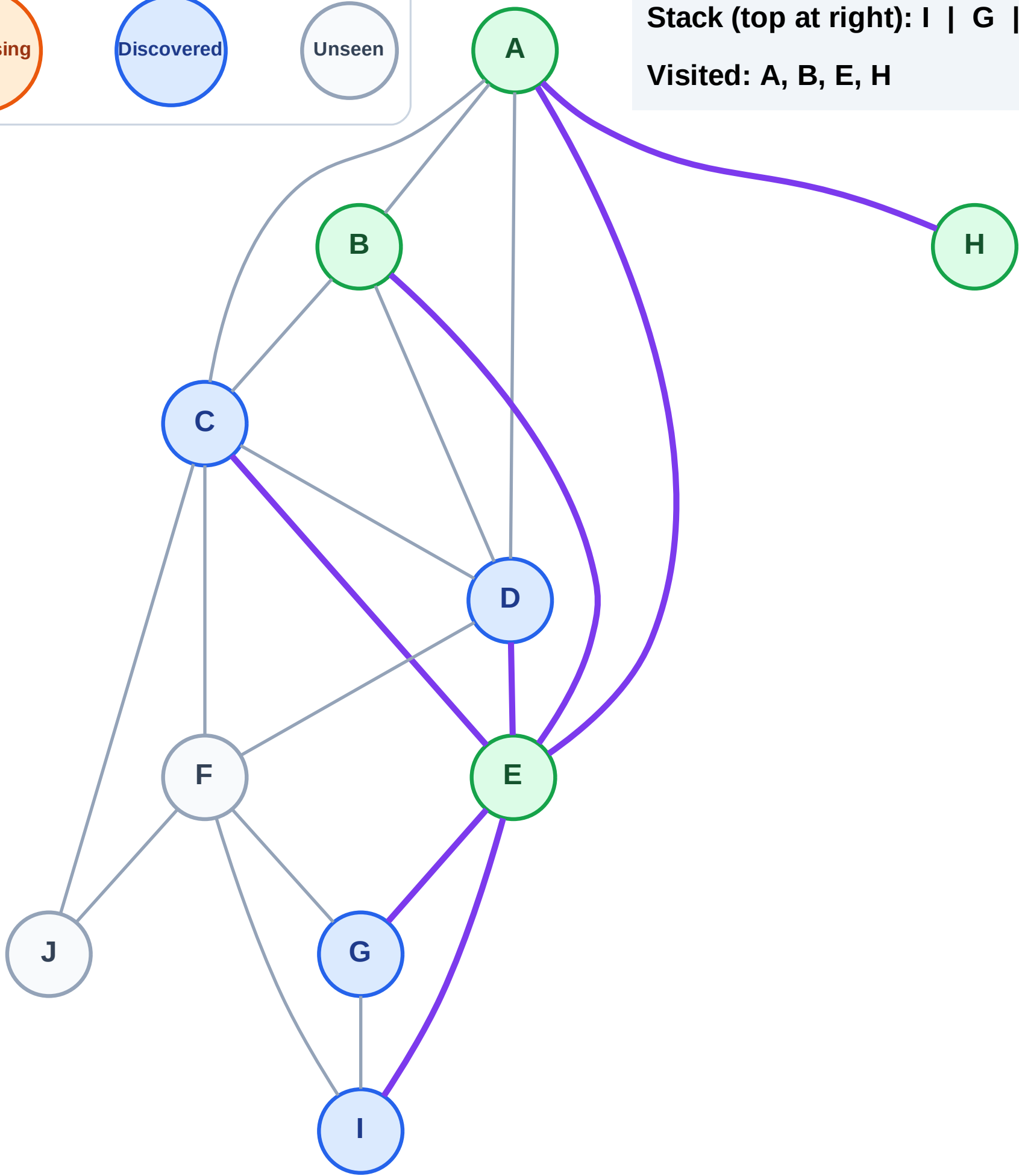
Step 9: No new neighbors from B

Legend



Stack (top at right): I | G | D | C

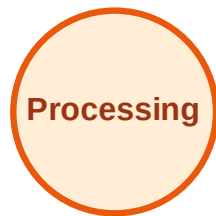
Visited: A, B, E, H



DFS Traversal

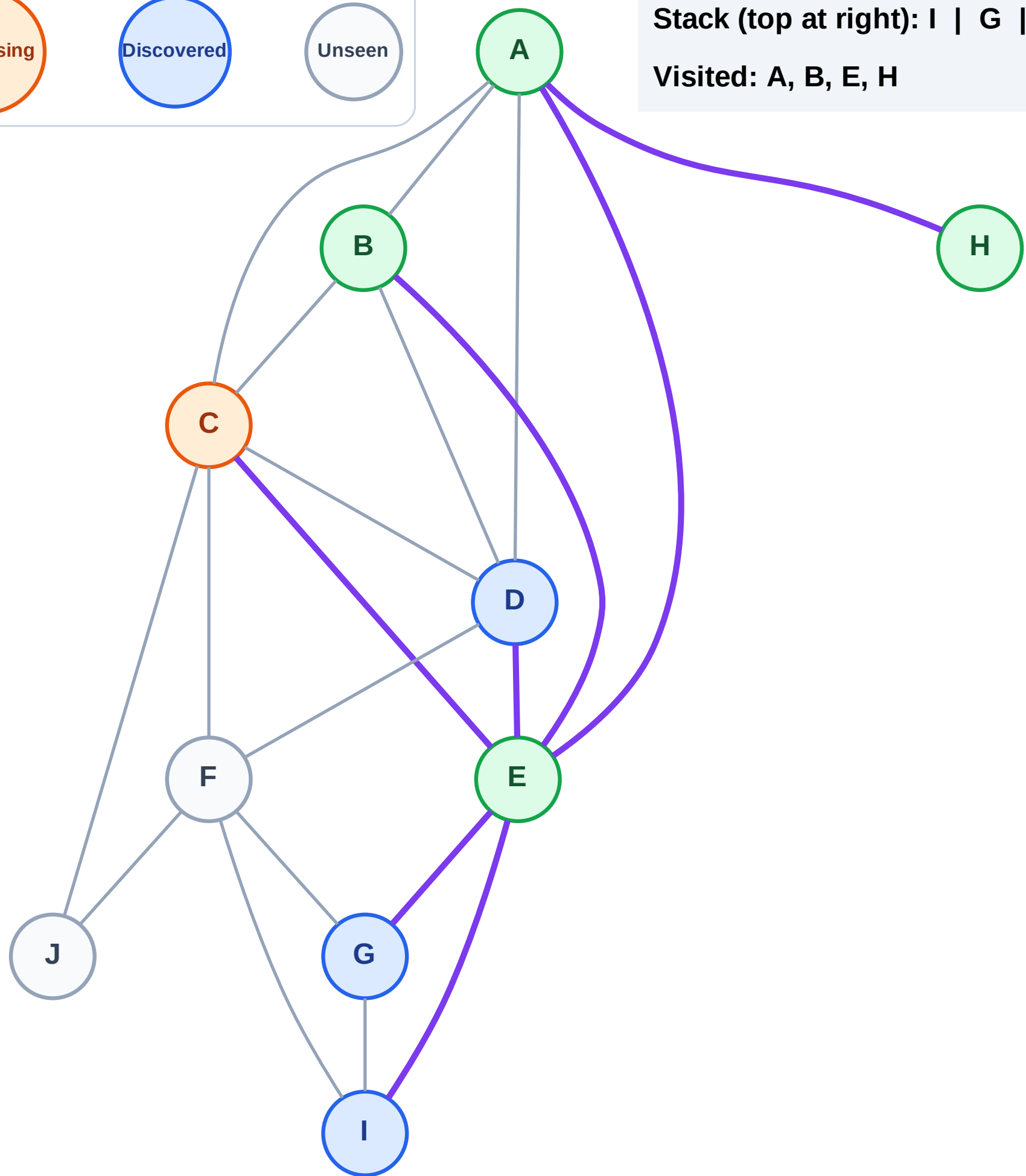
Step 10: Process node C

Legend



Stack (top at right): I | G | D

Visited: A, B, E, H



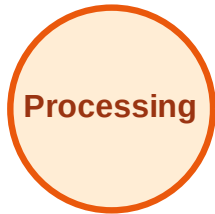
DFS Traversal

Step 11: Discover J, F from C

Legend



Complete



Processing



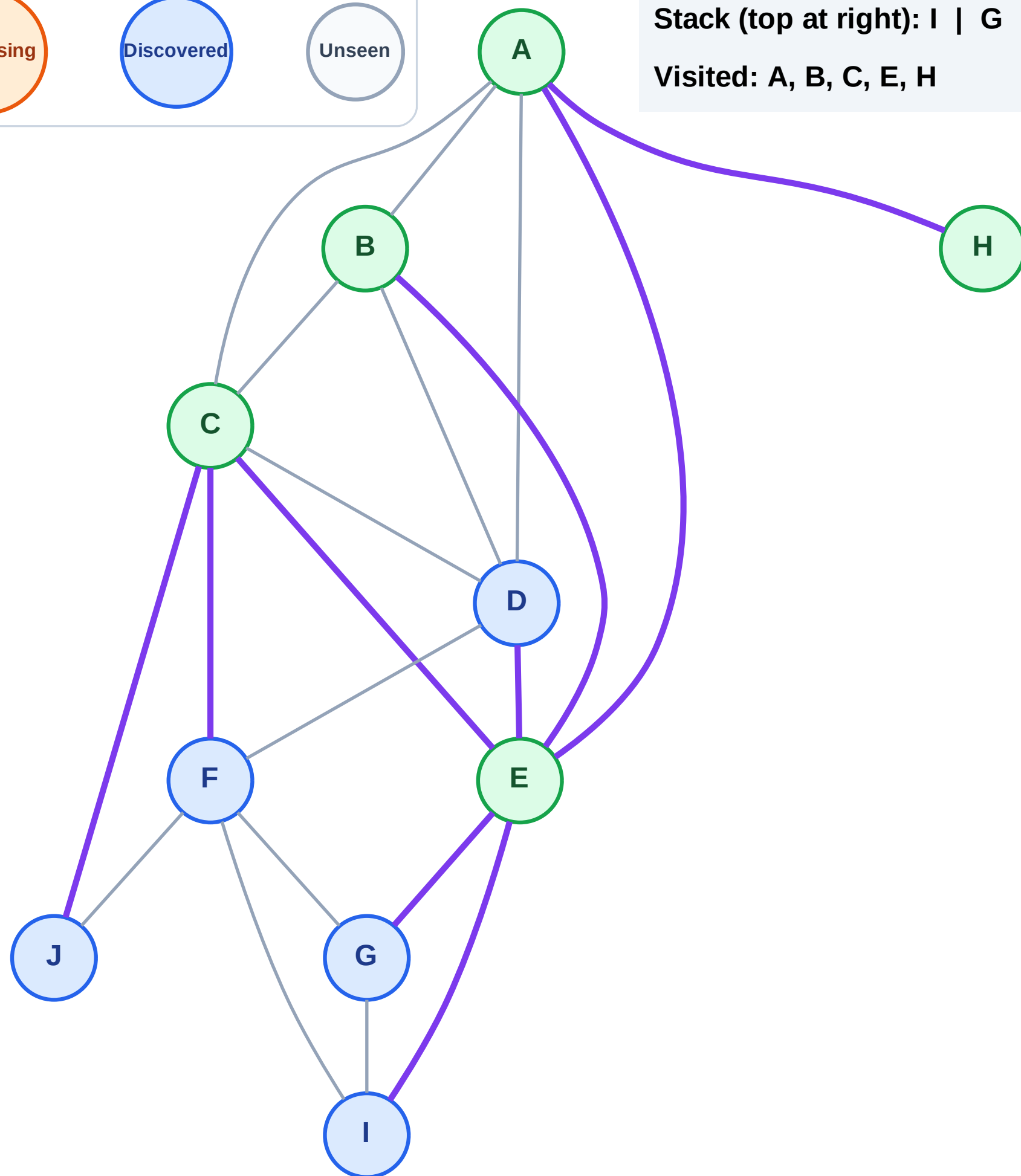
Discovered



Unseen

Stack (top at right): I | G | D | J | F

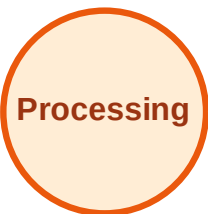
Visited: A, B, C, E, H



DFS Traversal

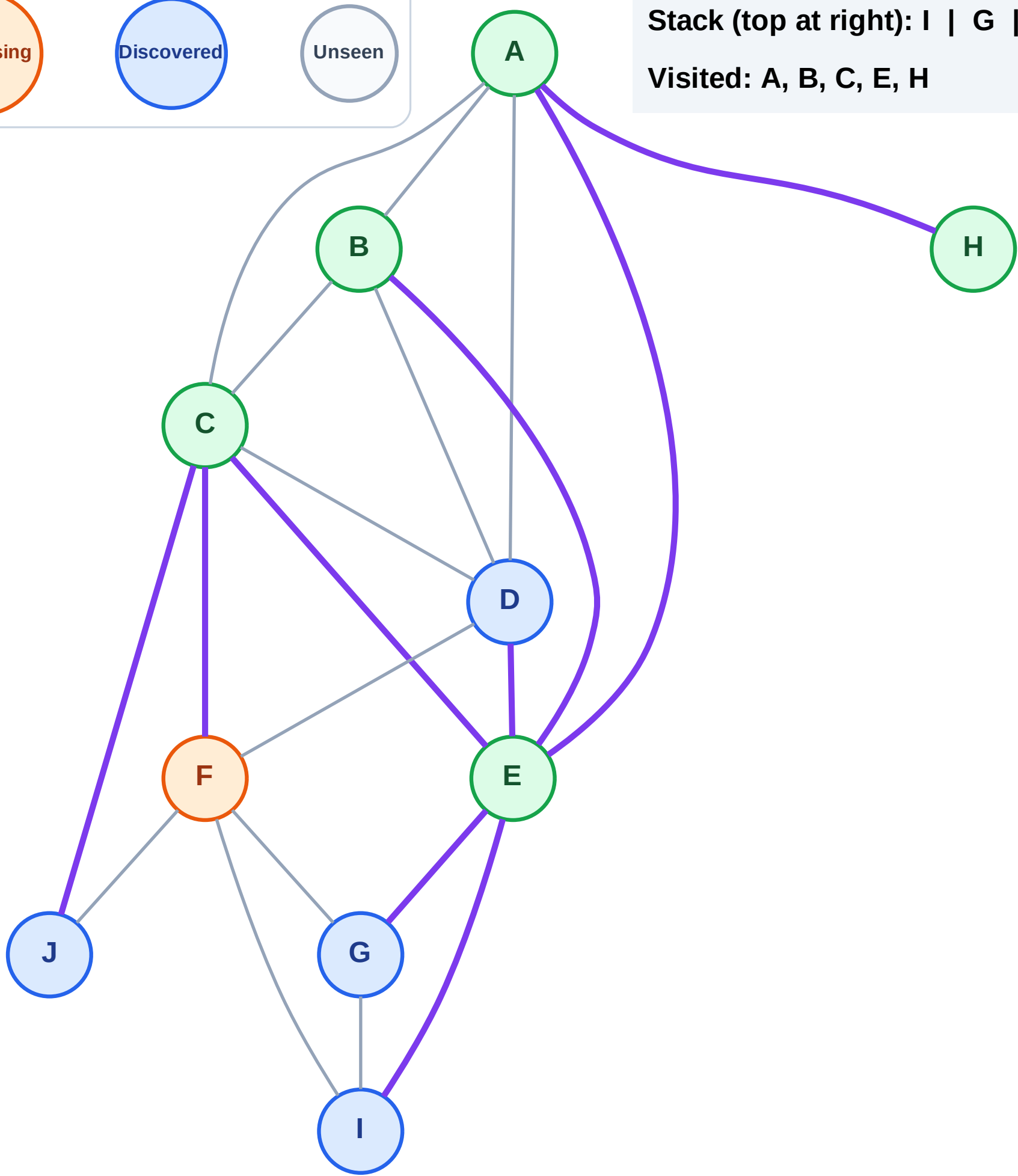
Step 12: Process node F

Legend



Stack (top at right): I | G | D | J

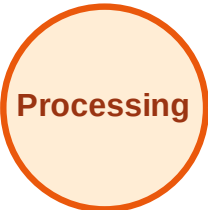
Visited: A, B, C, E, H



DFS Traversal

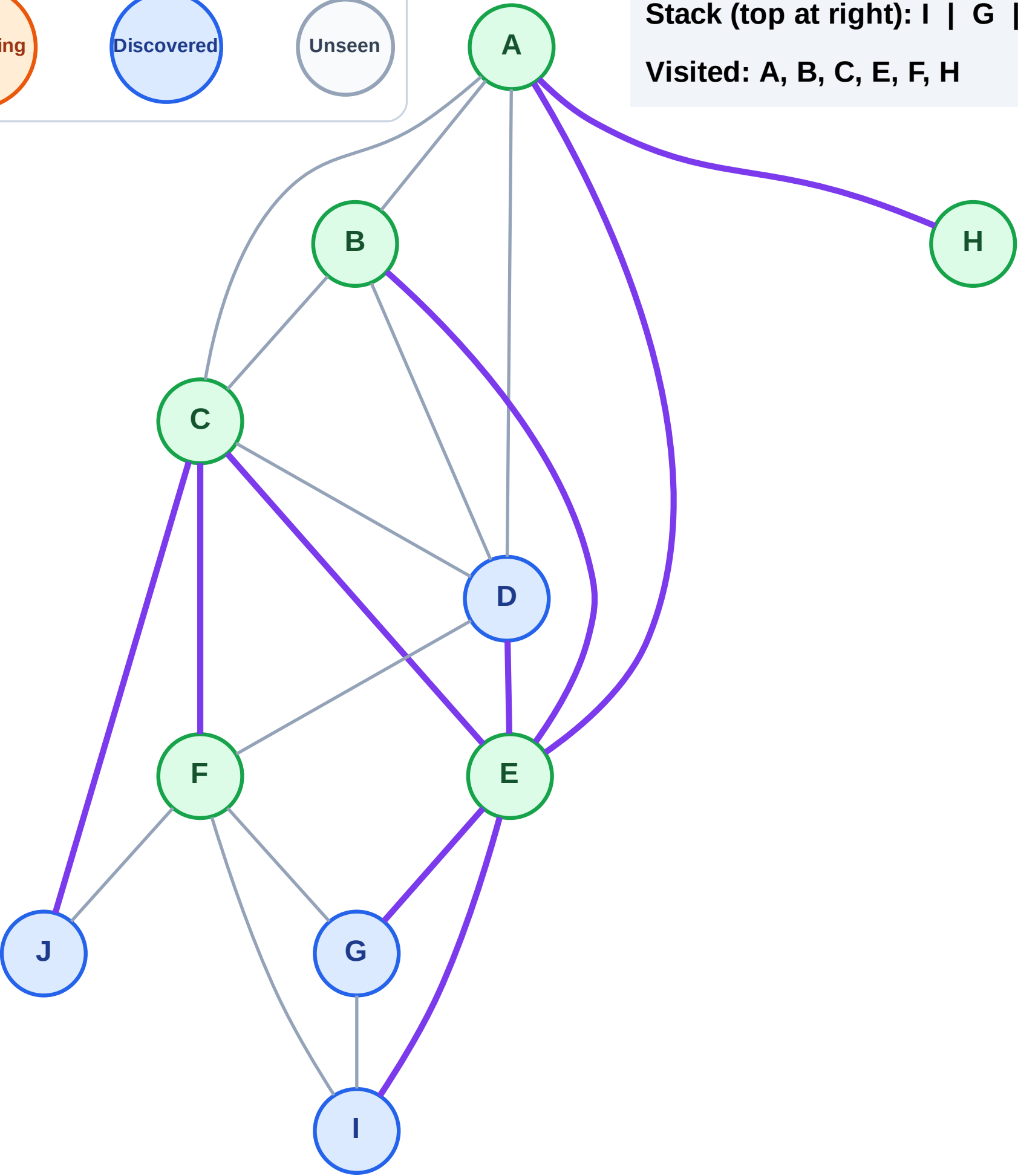
Step 13: No new neighbors from F

Legend



Stack (top at right): I | G | D | J

Visited: A, B, C, E, F, H



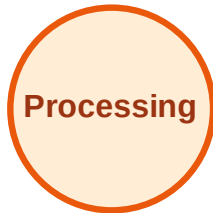
DFS Traversal

Step 14: Process node J

Legend



Complete



Processing



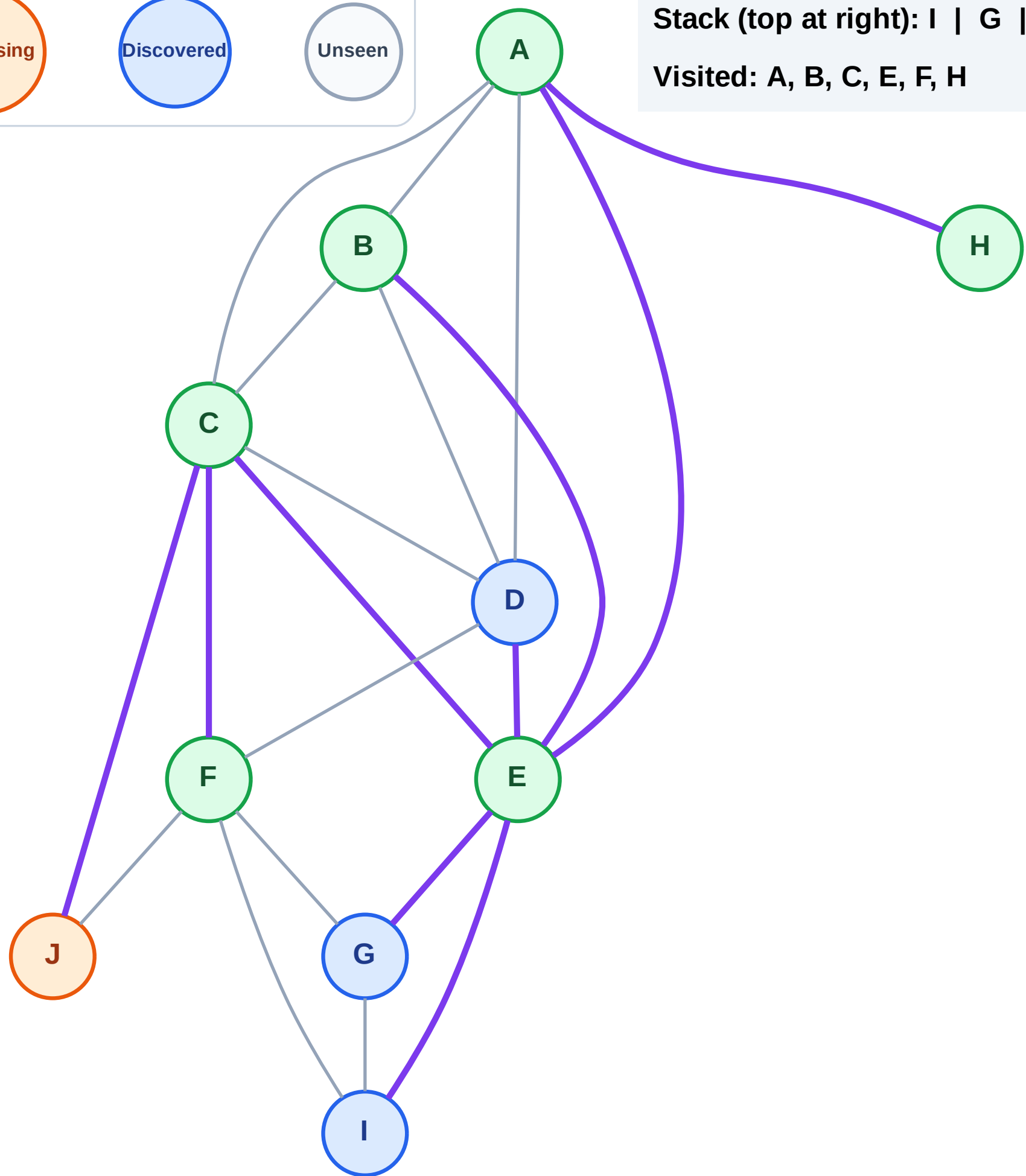
Discovered



Unseen

Stack (top at right): I | G | D

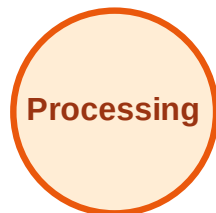
Visited: A, B, C, E, F, H



DFS Traversal

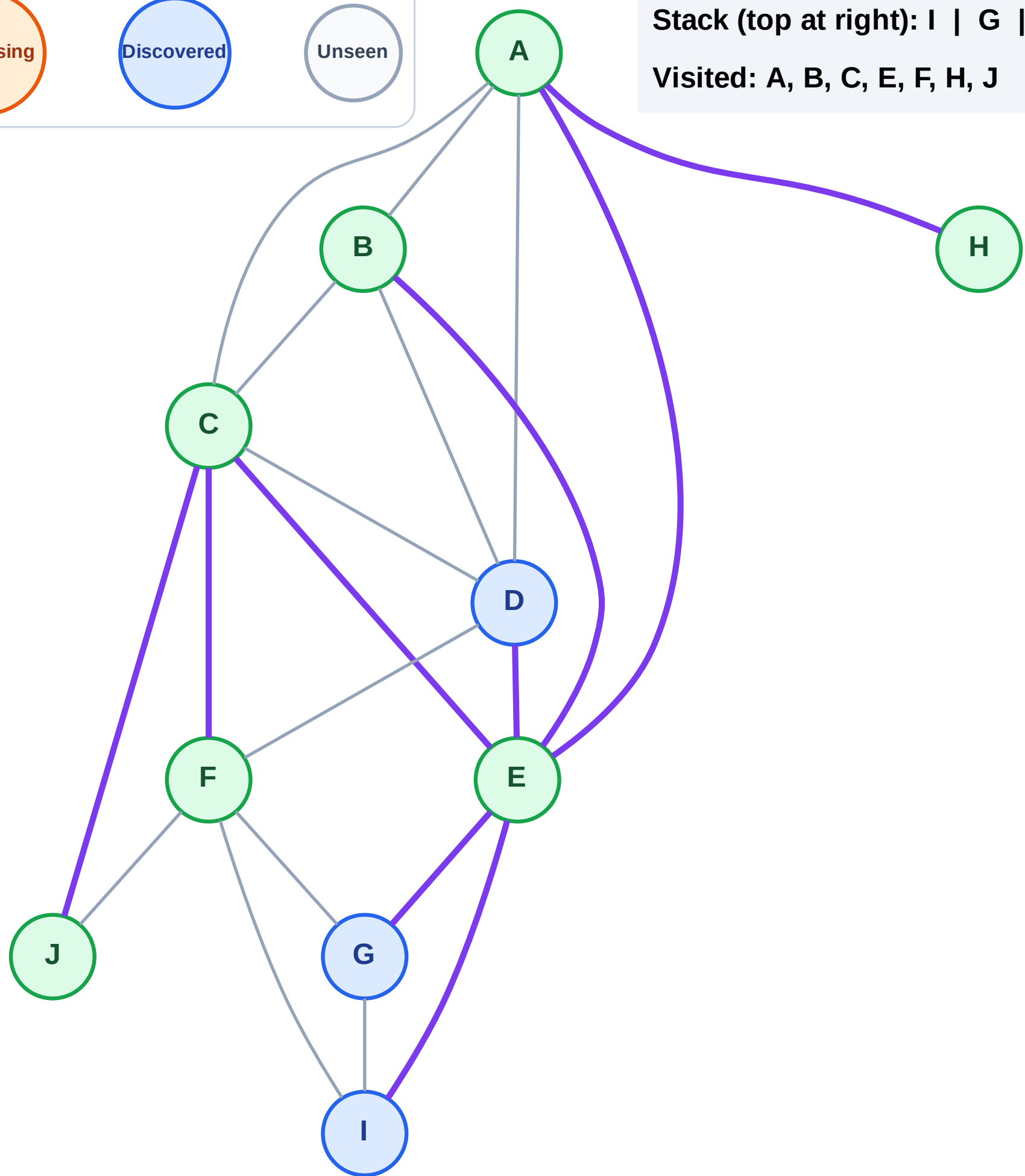
Step 15: No new neighbors from J

Legend



Stack (top at right): I | G | D

Visited: A, B, C, E, F, H, J



DFS Traversal

Step 16: Process node D

Legend

Complete

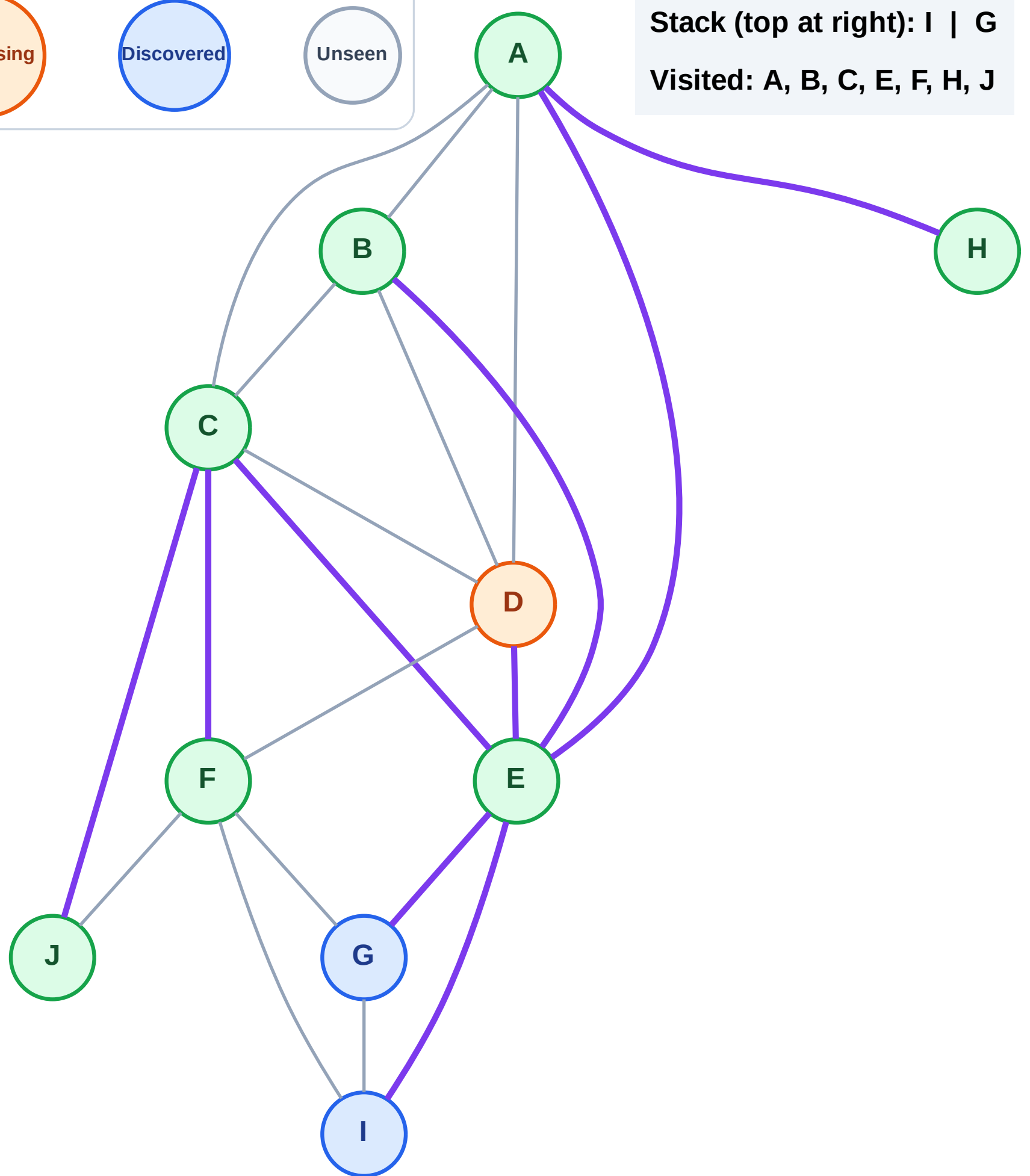
Processing

Discovered

Unseen

Stack (top at right): I | G

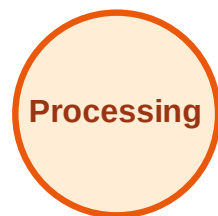
Visited: A, B, C, E, F, H, J



DFS Traversal

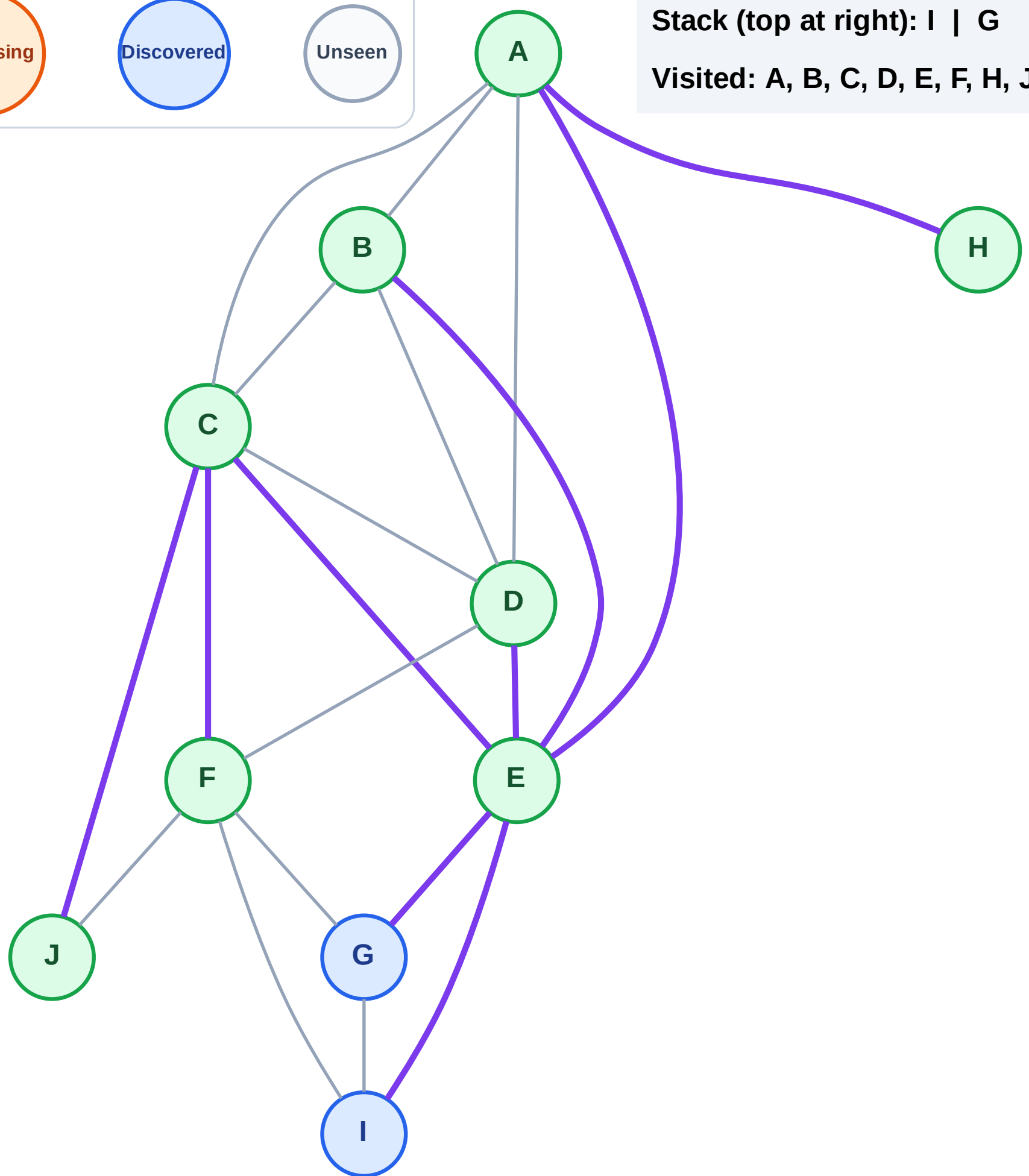
Step 17: No new neighbors from D

Legend



Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node G

Legend

Complete

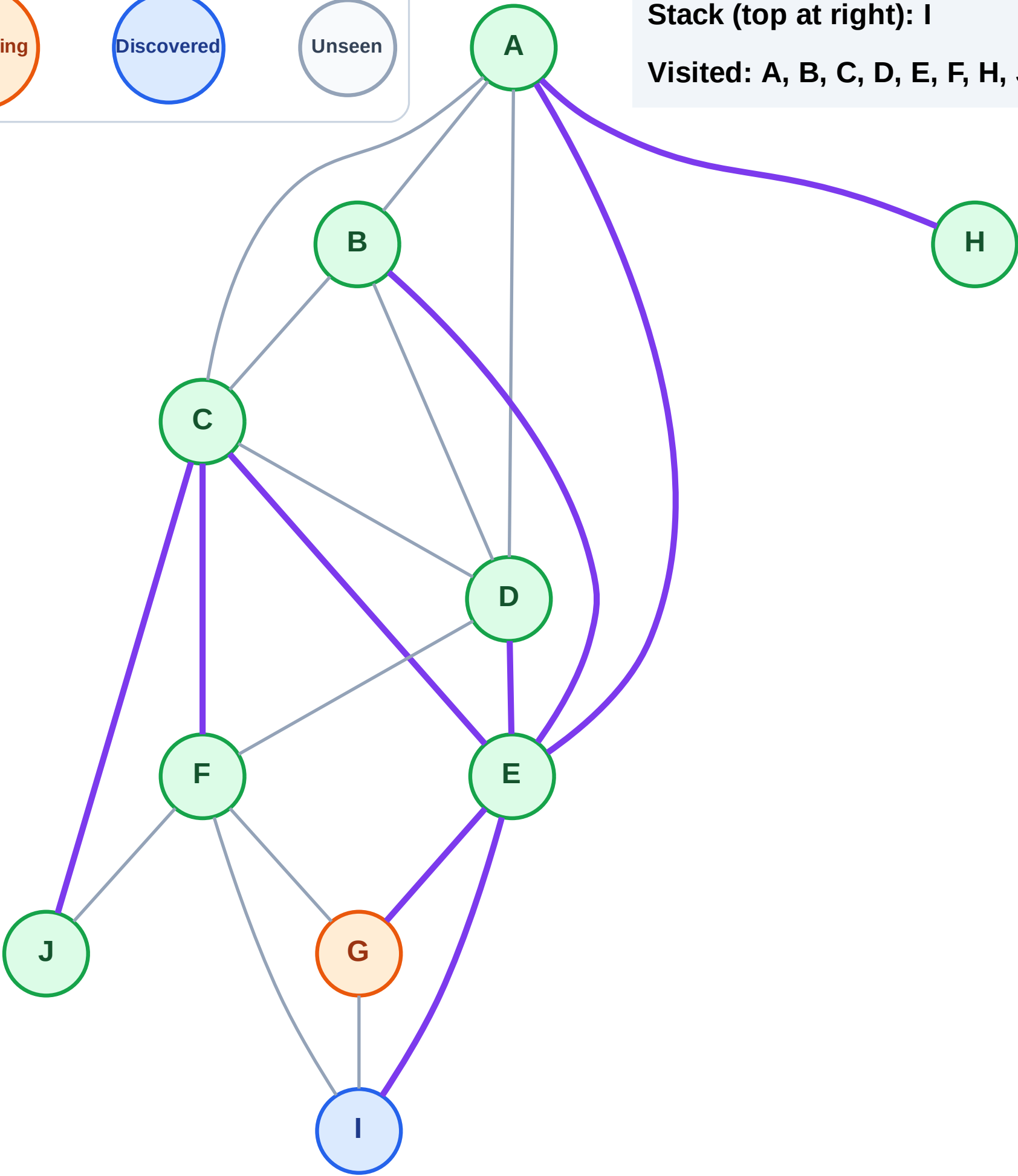
Processing

Discovered

Unseen

Stack (top at right): I

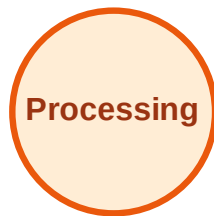
Visited: A, B, C, D, E, F, H, J



DFS Traversal

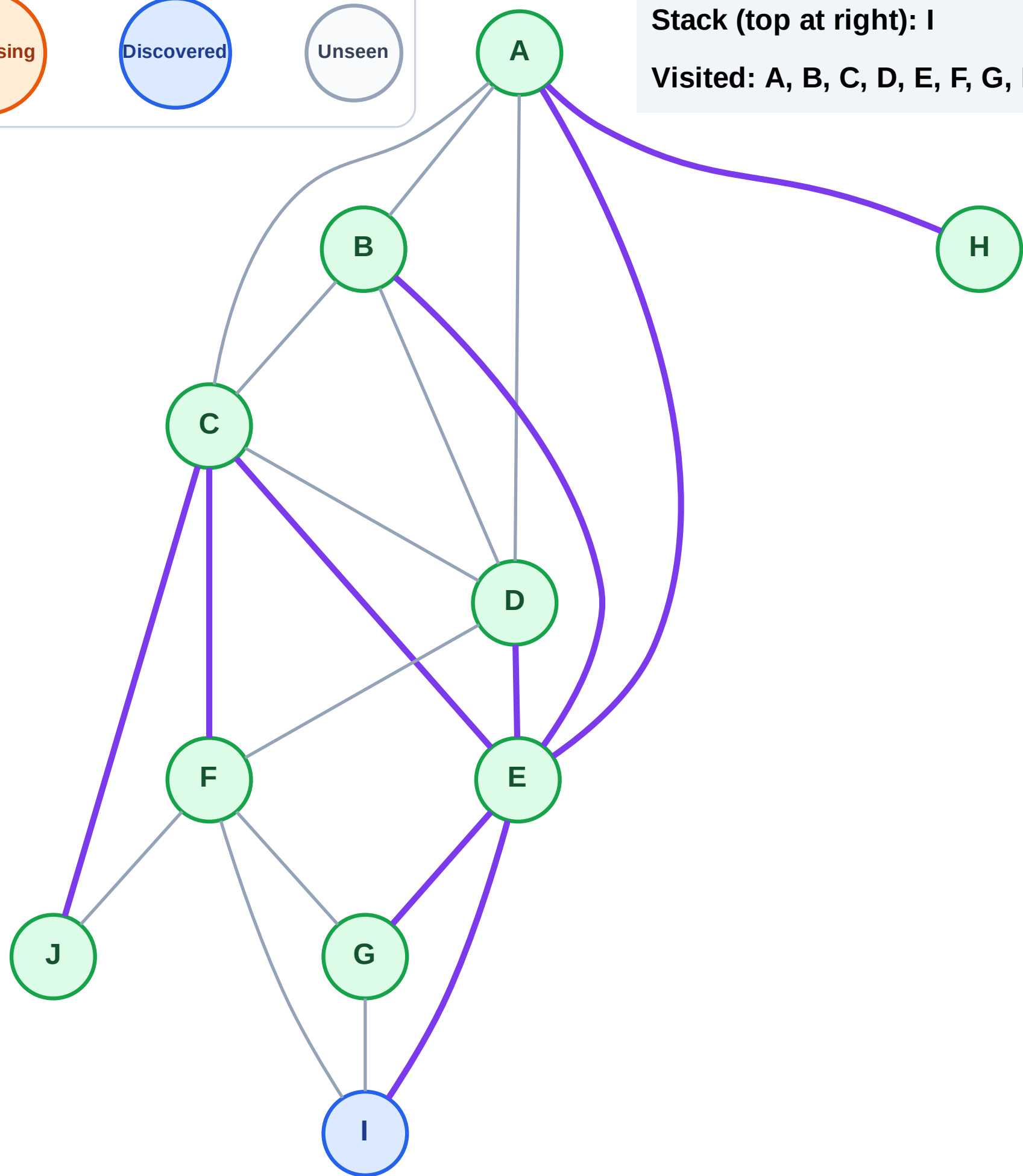
Step 19: No new neighbors from G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

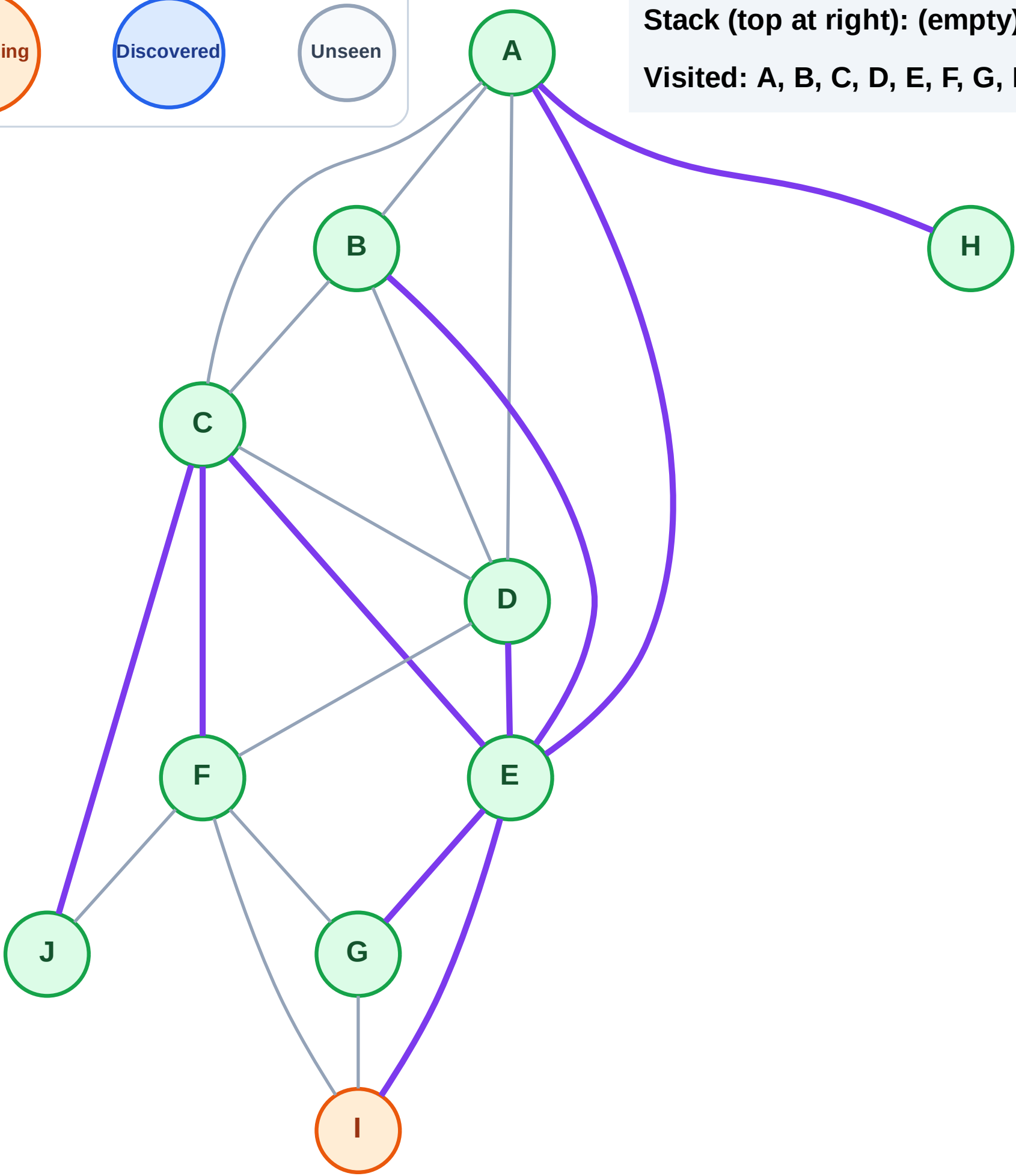
Step 20: Process node I

Legend



Stack (top at right): (empty)

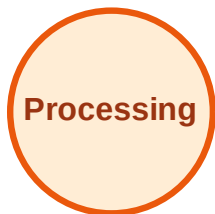
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

